

Domestic Outsourcing and Employment Security

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Abstract

Domestic outsourcing is known to reduce worker wages, but its effect on employment security—a key dimension of job quality—has not been studied. Using Brazil’s comprehensive employee-employer linked data, we find that outsourcing reduces exit from formal employment among cleaners and security guards during their first few years of tenure. The observed reduction in employment hazard is larger in cities with greater volatility in labor demand. The reduction is not attributable to differences in worker characteristics or differential exposure to local labor market shocks. The estimates suggest that outsourcing had larger positive effects on the net present value of worker earnings than implied by wage differentials alone. The patterns are consistent with a search-theoretic model in which outsourcing eases reassignment across firms.

Keywords: outsourcing, employment security, job search

JEL: J62, J63, L24

*First posted draft: November 2022. Emails: njguo@hku.hk, duoxili@bu.edu, mbwong@hku.hk. Access to Brazil’s RAIS database is governed by the Data Use Agreement between MIT and Brazil’s former Ministry of Labor. We thank Youzheng Jia, Zhongji Wei, and Lingrui Zhou for their excellent research assistance. We thank David Atkin and Mayara Felix for procuring MIT’s access to the database, and especially Mayara Felix for de-identifying, harmonizing, and translating the RAIS datasets, as well as discussions on related projects. We thank David Autor, Brigham Frandsen, Bob Gibbons, Simon Jäger, Kevin Lang, Jin Li, Johannes Schmieder, John Van Reenen, Birger Wernerfelt, Sammy Young, and seminar participants at HKU, MIT, and AASLE 2023 for helpful comments and suggestions. We acknowledge funding from the National Science Foundation, the Kuok Foundation, and the George and Obie Shultz Fund at MIT.

1 Introduction

To focus on their core competencies, firms have increasingly relied on outsourced workers for services once performed by direct employees—such as cleaning, security, IT, and HR. This paper explores how the rise of domestic outsourcing affects worker employment security and formal sector attachment. On one hand, outsourcing may exclude workers from firm-level benefits, including long-term employment security. On the other hand, professional service firms that operate large internal labor markets may prevent unemployment through flexible reassignment of workers across client firms. Thus, domestic outsourcing could redress the high levels of turnover and informality in developing labor markets (Ulyssea 2020; Donovan, Lu and Schoellman 2023). Yet, few studies have directly measured its effects on employment security.

In this paper, we study the effects of non-core activity outsourcing on worker employment security. We leverage Brazil’s comprehensive employment record and focus on cleaners and security guards, two major occupations for which outsourcing status is easily identified from industry codes. We define *employment security* as the inverse probability of involuntary exit from formal employment. We first estimate the effects of domestic outsourcing on employment security. We then investigate why the employment security of direct-hire workers appears to be lower than that of comparable outsourced workers.

For context, we begin with estimates of outsourcing wage differentials. We use a regression specification that includes controls for worker characteristics and fixed effects, following Dube and Kaplan (2010). Our wage estimates are broadly consistent with recent literature, which suggests that outsourcing reduces worker wages for low-wage workers, but less so for high-wage workers (Dube and Kaplan 2010; Goldschmidt and Schmieder 2017; Spitze 2022; Drenik et al. 2023). In the low-wage occupation of cleaners, outsourcing is associated with wages that are 11 percent lower. For the more professionalized and higher-wage occupation of security guards, outsourcing is associated with wages that are only 1.3 percent lower. Following Card, Heining and Kline (2013), we use event study designs to confirm that these outsourcing wage differentials reflect the causal effect of outsourcing rather than sorting effects.

Our main analysis then estimates the effects of outsourcing on the hazard of exit from formal

employment—which captures transitions both to unemployment and to informal employment—using a linear probability model, with controls for observable worker characteristics and occupation-region-year fixed effects. For robustness, we add controls for worker unobserved ability as measured by the two-way wage decomposition proposed by [Abowd, Kramarz and Margolis \(1999\)](#). We also account for unobservable worker selection by investigating regression coefficient stability in response to adding controls of observable characteristics, following the method proposed by [Oster \(2019\)](#).

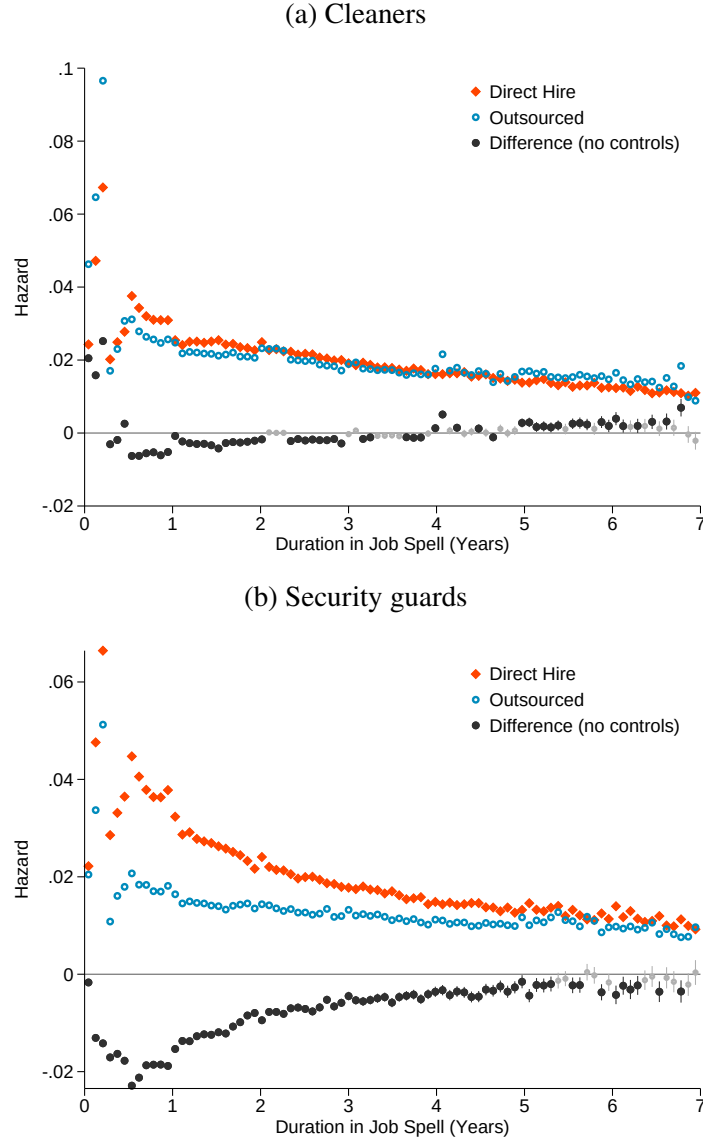
We find that outsourcing is robustly associated with reduced exit from formal employment. As shown in Figure 1, outsourced security guards have lower rates of involuntary transition from formal employment than direct-hire guards for the first six years of their employment spells. During the first year of employment spells, the predicted hazard of direct-hire security guards is roughly 50 percent higher. It is only in the seventh year of employment that their predicted hazard rates become the same. For cleaners, the hazard rates of direct-hire workers are higher than those of outsourced workers during the first three years of employment, but fall below those of outsourced workers thereafter. These differences are not explained by local labor market conditions or worker selection.

What drives these estimates? Outsourcing appears to reduce employment hazard more in cities with greater volatility. Specifically, we measure labor demand volatility using monthly changes in unemployment rates in six large metros. The observed reductions in employment hazard are larger in cities with more volatile labor demand. This evidence is consistent with the hypothesis, emphasized by prior literature, that outsourcing helps firms overcome labor demand fluctuations by enabling easier reassignment of workers across firms ([Abraham and Taylor 1996](#); [Houseman 2001](#); [Battiston, Espinosa and Liu 2021](#)).

The relationship between wages and city-level unemployment is also less negative for outsourced workers, even after controlling for worker fixed effects. This finding suggests that the wages of outsourced workers do not fall more in response to negative labor demand shocks. This finding thus casts doubt on the hypothesis that outsourcing reduces employment hazard by enabling firms to reduce wages rather than lay off workers in response to labor demand shocks.

To interpret our results, we develop a model in which outsourcing smooths demand fluc-

Figure 1: Hazard from Formal Employment, Direct-hire and Outsourced Workers



Notes: Each panel plots the raw probability of involuntary exit from formal employment during each 30-day interval, separately for outsourced and direct-hire workers, conditional on being employed at the beginning of that interval. The sample includes the first full-time spells at each employer between 2003-2010. The black and gray dots plot the estimated effect of being outsourced on involuntary exit from formal employment using the linear probability model in Equation (3), without any controls. Vertical bars show 95% confidence intervals.

tuations, affects productivity, and alters wage bargaining. Our search-theoretic model features match-specific productivity, wage bargaining, and endogenous separations. In the model, outsourcing alters employment relationships by allowing the worker to be reassigned to another firm when a negative productivity shock hits. Outsourcing may also alter the bargaining power

of workers. The model additionally allows outsourcing to affect initial match productivity—for example, through the hiring and matching expertise of intermediaries, positive selection of outsourced workers, or training provided by intermediaries—which could independently lower hazard rates for outsourced workers. The model predicts that the hazard rate exhibits a “crossing” pattern—wherein the hazard rate of outsourced workers is lower than that of directly employed workers during the early period of employment spells, but becomes higher than that of direct-hire workers as time progresses—if and only if reassignments across firms are sufficiently frequent.

Finally, we estimate the net present value (NPV) of earnings for outsourced and direct-hire workers, accounting for both employment-to-unemployment (E-U) and employment-to-employment (E-E) transitions. When considering only wage differentials, the NPV of outsourced workers is lower than that of direct-hire workers: 88.5% for cleaners and 96.3% for guards. However, incorporating differences in E-U hazard rates substantially narrows this gap, as outsourced workers face lower rates of involuntary separation and recover more rapidly from layoffs. When we additionally account for E-E transitions—which occur more frequently for outsourced workers and lead to wage growth—the NPV of outsourced workers exceeds that of direct-hire workers in both occupations, reaching approximately 101.3% of direct-hire NPV for cleaners and 102.8% for guards. We conclude that the negative effect of outsourcing on worker earnings through lower wages is substantially offset by improvements in employment security.

The contribution of this paper is threefold. First and most importantly, this paper is the first to estimate the effects of domestic outsourcing on job hazards using administrative employment records (see surveys by [Davis-Blake and Broschak 2009](#); [Bernhardt et al. 2016](#)). Prior evidence on outsourcing and job security is limited to [Batt, Doellgast and Kwon \(2005\)](#); [Batt, Holman and Holtgrewe \(2009\)](#), who use a small survey sample to document lower *perceived* job security among outsourced call center workers. Our findings suggest that outsourcing improved *actual* employment security in Brazil. In closely related work, [Felix and Wong \(2025\)](#) study an earlier period in Brazil and estimate that outsourcing legalization led to employment and reallocation effects consistent with reduced labor market frictions, but lack statistical power to examine worker transitions. We complement their work by estimating the effects of outsourcing on employment security in two large occupations and by documenting that these effects are larger

in more volatile labor markets. Our interpretive framework features flexible reassignment of workers across firms, a channel absent from prior models of outsourcing (Spitze 2022; Bilal and Lhuillier 2021).

Second, we contribute novel estimates of outsourcing wage differentials from a developing country. Most existing estimates come from the US (Abraham 1990; Dube and Kaplan 2010; Spitze 2022), the UK (Berlinski 2008), and Germany (Goldschmidt and Schmieder 2017). We confirm the importance of firm-level wage premia and find that the wage effects of outsourcing are more negative in a low-wage occupation.

Third, we contribute novel estimates of the net present value of employment under outsourcing. By integrating wage differentials with differences in E-U and E-E transition rates, we show that the NPV gap between outsourced and direct-hire workers is substantially smaller—and in some specifications reversed—relative to what wage differentials alone would imply.

Our work is related to, but distinct from, two other bodies of literature. A strand of the literature focuses on *core* activity outsourcing, which is typically motivated by a desire to circumvent labor regulation (Estefan et al. 2024; Bertrand, Hsieh and Tsivanidis 2021; Jiménez and Rendon 2022). We focus on *non-core* activity outsourcing, where task-specific economies of scale in human resources may enable flexible reassignment. Another strand studies fixed-term contracts in Europe, which are associated with *reduced* employment security (Blanchard and Landier 2002; Bentolila and Saint-Paul 1992; Cahuc and Postel-Vinay 2002; Daruich, Addario and Saggio 2020). We find a different pattern in Brazil, where indefinite employment in specialized service firms appears to ameliorate the high turnover that characterizes direct-hire employment in a developing economy context.

The rest of the paper is organized as follows. Section 2 provides background. Section 3 estimates outsourcing wage differentials. Section 4 estimates the effects of outsourcing on employment security. Section 5 presents an interpretive framework. Section 6 estimates the net present value of outsourced and direct-hire employment. Section 7 concludes.

2 Background

2.1 Data and Sample Construction

We use Brazil’s employee-employer matched administrative data, *Relação Anual de Informações Sociais* (RAIS), which cover the near universe of Brazil’s formal-sector workers. The RAIS data include annual information on the start and end dates of each employment spell, the average monthly wage over that period, and several demographic variables (such as education, gender, race, and age), which are collected through a mandatory survey administered by the Brazilian Ministry of Labor and Employment. These data are of high quality, since firms are fined for failure to report and workers cannot receive government benefits unless accurate information is reported.

We focus on data from 2003 to 2010, a period that is uncontaminated by the effects of Brazil’s 1993 outsourcing legalization and has both consistent occupation codes and exact start and end dates for employment spells. To identify direct-hire and outsourced workers, we use detailed industry and occupation codes.

Despite their richness, these data have two limitations. First, we do not observe worker-firm-intermediary linkages, so we cannot characterize the match between outsourced workers and client firms.¹ Second, there is a substantial informal sector in Brazil. Only 68% and 79% of cleaners and security guards, respectively, are in the formal sector and hence covered in our sample (Appendix Table A.4). Missing observations in our data could represent either unemployment or informal employment. We address this shortcoming by leveraging information on separation reasons.

For our hazard estimation, we construct employment histories for individual workers as follows. We restrict attention to workers aged 18-65 in full-time jobs (at least 35 hours per week) and exclude workers with temporary contracts.² We say that an employment spell ends in an *exit from formal employment* if there is more than one week between the spell’s end and

¹This is a problem that plagues most administrative employment records with only one known exception (Drenik et al. 2023).

²These contracts are uncommon and subject to approval by the Ministry of Labor (MTE) to meet temporary increases in demand. Many of these contracts last for three months.

the start of the worker’s next full-time employment spell. We count exits as *involuntary* if the spell did not end in retirement, death, or quitting, so severance compensation must be made. Appendix Figure C.4 shows that our main results remain highly similar when quits are included instead. Appendix A provides detailed data definitions.

2.2 Our Focus: Cleaners and Security Guards

We focus on cleaners and security guards for two reasons. First, both are large occupations where a substantial number of workers are employed by contract firms and within which the task requirements are relatively homogeneous. Second, there is a clear mapping from industry codes to contract firm status. In other occupations, the same industry code may include both intermediate services to client firms and final services to consumers, preventing a clean identification of outsourcing status.³ We therefore confidently identify outsourced workers using detailed industry and occupation codes in our data. These classifications are presented in Appendix A.

The comparison between these two occupations provides insights into how the effects of outsourcing may be heterogeneous across occupations. Security guards are highly professionalized, regulated, and well-paid. Because of high crime rates and inadequate public provision of policing, security guards in Brazil undergo mandatory training administered by the Brazilian government and face regulatory requirements for gun carry licenses. By contrast, cleaners are an unlicensed occupation in Brazil. They are also the lowest-paid occupation in the formal sector. The mean monthly wage of cleaners in 2010 is roughly equal to one-half of the mean monthly wage of security guards. Appendix Table A.5 shows that the outsourcing firms in our sample are large, specialized, and well-established: the median outsourced worker’s firm employs over

³We investigated several other occupations and found that none permits a clean identification of outsourcing status. General drivers (CBO 98510) in the road transport industry include public transportation workers who provide final services to consumers. Truck drivers (CBO 98560) in road transport include freight carriers serving end customers, and direct-hire truck drivers in industries such as construction or agriculture perform tasks that differ substantially from those at transport intermediaries. Cooks (CBO 53110) in food services include both restaurant workers and catering workers providing intermediate services to client firms (e.g., corporate cafeterias), and the industry code does not distinguish between the two. Telemarketing operators (CBO 38025) are a significantly smaller occupation and split between intermediaries and telecommunications companies. Office assistants and building receptionists have less than 10% of employment in service intermediaries. See also Dube and Kaplan (2010), who use the same identification strategy for cleaners and security guards in U.S. data.

Table 1: Summary Statistics of Employment Spells, Brazil, 2003-2010

	Cleaners		Security guards	
	Direct-hire	Outsourced	Direct-hire	Outsourced
Worker characteristics at spell start:				
Age	32.4 [10.1]	33.1 [9.81]	36.1 [10.8]	32.6 [7.93]
Years of schooling	7.82 [3.25]	7.37 [3.07]	8.53 [3.53]	9.88 [2.79]
Male	0.50 [0.50]	0.43 [0.50]	0.96 [0.21]	0.94 [0.24]
Non-white	0.45 [0.50]	0.51 [0.50]	0.52 [0.50]	0.50 [0.50]
Contract hours	43.6 [1.49]	43.7 [1.29]	43.3 [2.12]	43.7 [1.52]
Share of spells that end within:				
one year	0.50	0.59	0.47	0.38
1-2 years	0.13	0.13	0.14	0.14
2-3 years	0.06	0.06	0.06	0.07
3-4 years	0.03	0.03	0.03	0.04
4-5 years	0.015	0.013	0.015	0.019
5-6 years	0.007	0.007	0.008	0.010
6-7 years	0.003	0.002	0.003	0.004
Share with unobserved spell end	0.26	0.18	0.28	0.33
Reason for spell end:				
Involuntary exit from formal sector	0.51	0.45	0.49	0.34
Voluntary exit from formal sector	0.13	0.15	0.08	0.08
Transition to other formal job	0.06	0.16	0.09	0.17
Other	0.05	0.06	0.06	0.07
Observations	2681874	1897053	797534	1201625

Notes: The sample is all employment spells of security guards and cleaners, respectively, between 2003 and 2010. Standard deviations are displayed in brackets.

1,500 workers and has been in operation for at least 15 years.

Table 1 shows the characteristics of the employment spells of outsourced and direct-hire workers, including age, education, gender, and race at spell start. Anticipating our main result below, the employment spells for outsourced cleaners and security guards are *less* likely to end in involuntary exit from the formal sector than direct-hire cleaners and security guards,

respectively.

3 Effect of Outsourcing on Wages

As a first look at the effects of outsourcing in these two occupations, we measure the effect of outsourcing on worker wages. Following [Dube and Kaplan \(2010\)](#), we estimate the following equation using yearly panels of security guards and cleaners, respectively:

$$\ln w_{it} = \gamma O_{it} + \theta_{omt} + X'_{it}\beta + \alpha_i + \epsilon_{imt}, \quad (1)$$

where t indexes year, i indexes the worker, w_{it} is the average real monthly wage, O_{it} indicates whether the worker is outsourced, θ_{omt} is a suboccupation-microregion-year fixed effect, $X'_{it}\beta$ are the effects of time-varying observable worker characteristics (such as education and age), α_i controls for individual fixed effects, and ϵ_{imt} is a composite error that may include idiosyncratic worker-firm match effects.⁴ We then check whether estimated outsourcing wage differentials have a causal interpretation by plotting event studies for workers who change contractual arrangements, following the method of [Card, Heining and Kline \(2013\)](#).

Outsourcing wage differentials may be attributable to either compensating differentials or differences in labor market rents. To understand the source of wage differentials, we investigate the extent to which outsourcing wage differentials are explained by differences in firm-level wage premia. We measure firm-level wage premia using the two-way decomposition proposed by [Abowd, Kramarz and Margolis \(1999\)](#) — henceforth AKM — as described in Appendix A.

We first confirm that the AKM decomposition provides a useful measure of firm-level wage premia for cleaners and security guards. After correcting for measurement error, we find that the AKM firm effects estimated using only cleaners and security guards, respectively, are very

⁴An alternative approach is to follow [Goldschmidt and Schmieder \(2017\)](#), who estimate wage differentials using “on-site outsourcing events.” We do not follow this approach for two reasons. First, Brazilian labor law prohibits nominal wage reductions through the firing and rehiring of workers at an intermediary to perform the same job. As a consequence, estimates of wage differentials using such events are likely to be biased upward. Second, as documented by [Felix and Wong \(2025\)](#), on-site outsourcing is exceedingly rare in Brazil. Given the rarity, wage differentials estimated using this method necessarily use a highly selected population of workers.

highly correlated with the AKM effect estimated for all other workers. If a firm pays 10% higher wages to other workers, it pays 6.1% higher wages to cleaners and 9.7% higher wages to security guards (Appendix Figure B.1). We then investigate whether firm-level wage premia explain the observed outsourcing wage differential.

Cleaners. Table 2 Panel A displays the estimated outsourcing wage differentials for cleaners, which is -11.0 log points in our preferred specification of Column (3). Column (1) shows that, with occupation-microregion-year fixed effects, the wages of outsourced cleaners are roughly 19.4 log points lower than direct-hire cleaners. With additional demographic controls in Column (2), the estimate changes very slightly to 17.8 log points. With added individual fixed effects, as in Column (3), the wages of outsourced cleaners are smaller at 11.0 log points, suggesting that there is some unobserved selection into outsourcing.

Column (4) shows that the outsourcing wage differential is much smaller after controlling the AKM firm effect, at 4.9 log points. Column (5) uses a split sample IV approach, to remove the influence of measurement error, with AKM firm effects estimated from two equally sized samples of workers that include neither cleaners nor security guards, to remove mechanical correlation arising from using the same data on both sides of the equation (following [Goldschmidt and Schmieder 2017](#)). The correlation between cleaner wages and AKM firm effects is only somewhat attenuated in this specification. These estimates suggest that differences in firm-level wage premia substantially explain the outsourcing wage differential.

Figure 2 Panels (a) and (b) show that cleaners who switch from direct-hire to outsourced jobs experience relative wage declines, while cleaners who switch from outsourced to direct-hire jobs experience relative wage increases. There are no significant pre-event trends during the two years prior, which implies that the estimated outsourcing wage differential is likely to capture the causal effect of outsourcing.

Security guards. For security guards, a higher-wage occupation, the outsourcing wage differential is very small. Our preferred estimate, which controls for unobservable worker heterogeneity, is -1.3 log points.

Table 2: Outsourcing Wage Differential, Brazil, 2003-2010

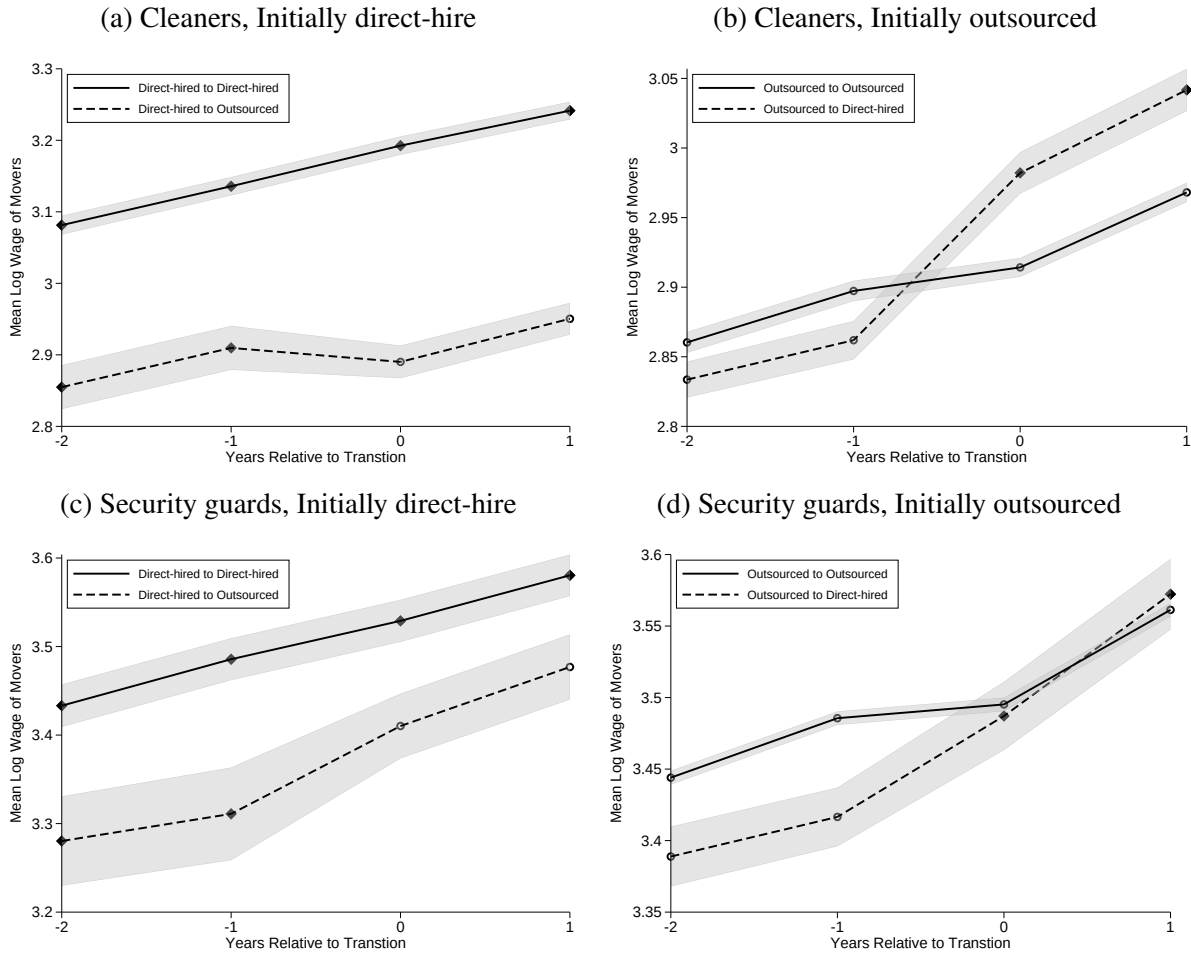
Dep. var.: Log real wage	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Cleaners</i>					
Outsourced	-0.194 (0.006)	-0.178 (0.006)	-0.110 (0.003)	-0.049 (0.002)	-0.072 (0.003)
AKM firm effect				0.413 (0.007)	0.332 (0.007)
Observations	7834355	7834355	6150003	6131898	5904192
R^2	0.30	0.35	0.94	0.94	0.04
<i>Panel B: Security guards</i>					
Outsourced	-0.173 (0.026)	-0.137 (0.020)	-0.013 (0.005)	-0.014 (0.003)	-0.003 (0.005)
AKM firm effect				0.690 (0.011)	0.397 (0.012)
Observations	4768878	4768878	4253501	4251537	4176299
R^2	0.44	0.48	0.93	0.93	0.03
Occ X Year X Microregion FE	X	X	X	X	X
Demographic controls		X	X	X	X
Worker FE			X	X	X

Notes: Sample includes all cleaners and security guards, respectively, observed at year-end in RAIS between 2003-2010. Demographic controls include a full set of race X gender X education dummies interacted with age, age squared, and age cubed. AKM firm effect is estimated from a wage regression using the full worker sample in column (4), and two non-overlapping random sets of workers in column (5). Standard errors are clustered at both worker and firm level, and displayed in parentheses.

Table 2 Panel B Column (1) shows that the wages of outsourced security guards are roughly 17.3 log points lower than direct-hire security guards with occupation-microregion-year fixed effects. With additional demographic controls in Column (2), the estimate is similar, at 13.7 log points. With the individual fixed effects, as in Column (3), the estimate is only 1.3 log points. Columns (4) and (5) show that the estimated outsourcing wage differential remains small after accounting for differences in firm-level wage premia.

Figure 2 Panels (c) and (d) show that security guards who move from direct employment to outsourcing and from outsourcing to direct employment both experience a wage increase. The asymmetry in wage responses suggests some degree of endogeneity in worker mobility, but

Figure 2: Wage Evolution, Job Switchers



Notes: This figure shows the mean log wages of workers who transitioned between establishments in 2003-2010. We restrict the sample to job switchers who are observed not to change establishments during the two years before and during the two years after the transition. Panels (a) and (c) show cleaners and security guards, respectively, who were initially direct employees and switched to outsourcing. Panels (b) and (d) show cleaners and security guards, respectively, who were initially outsourced and switched to direct employment.

confirms that the outsourcing wage differential is likely to be small.

Appendix Figure B.2 and Table B.1 show that the negative wage effects of outsourcing increase with tenure, for both cleaners and security guards, even after controlling for demographic variables such as age.⁵

⁵Our analysis focuses on wages. Appendix Table B.3 examines non-wage benefits using PNAD data. The evidence on differences in non-wage benefits between outsourced and direct-hired workers is mixed and does not indicate a consistent negative relationship with outsourcing.

4 Effect of Outsourcing on Exit from Formal Employment

Having examined the wage effects of outsourcing, this section estimates the effect of outsourcing on the rate at which workers exit from formal employment. These estimates are important for understanding the welfare effects of domestic outsourcing on workers. If outsourcing facilitates their transitions across firms in response to demand fluctuations, then outsourcing should reduce hazards from formal employment and thereby benefit workers. To our knowledge, this paper is the first to investigate these effects.

4.1 Method

We construct the first full-time spell for each worker at each employer. We estimate the hazard function only at duration less than or equal to 7 years because of small sample sizes with longer duration. We censor spells ending in a job-to-job transition, in which case we do not know when the spell would have ended in exit from formal employment.

Following [Schmieder and Trenkle \(2020\)](#), we estimate the hazard rate of exit from formal employment at each month $\tau = 1, \dots, 84$ using the following regression model:

$$y_{i\tau} = \alpha_\tau + \delta_\tau O_i + X_i' \beta_t + \theta_{omt} + \epsilon_{i\tau} \mid \tau_i \geq \tau, \quad (2)$$

where τ_i is the month when individual i exits formal employment. In each regression, conditional on worker i has survived in an employment relationship for $\tau - 1$ months, the dependent variable $y_{i\tau}$ is a dummy indicating whether worker i exited formal employment at month τ . O_i indicates whether worker i is outsourced. X_i are worker-level controls. θ_{omt} are suboccupation-microregion-year fixed effects. Estimating Equation (3) at each τ provides a vector of α_τ which represents the average rate of hazard from formal employment of direct-hire workers in month τ , while δ_τ represents the shift in the hazard rate of outsourced workers in that month, which measures the effect of outsourcing on exit from formal employment.

There are two main potential sources of bias in our hazard estimates. First, outsourced workers may be more prevalent in certain locations and therefore are differentially exposed to

local macroeconomic fluctuations. To account for this confounding influence, we add flexible controls for microregion-suboccupation-year fixed effects.

Second, outsourced workers may be selected. To address this concern, we use a rich set of demographic controls, including gender, age, race, and education at the start of the employment spell, as well as AKM worker effects as a proxy for unobserved worker ability. We also assess the potential impact of unobserved worker selection using the method proposed by [Oster \(2019\)](#). Specifically, we investigate how coefficient stability is affected by the addition of worker-level controls, after controlling for microregion-suboccupation-year fixed effects. Following [Oster \(2019\)](#), we plot bias-adjusted estimates β^* under the benchmark assumption $\delta = 1$, meaning that selection on unobservables—after partialing out the microregion-suboccupation-year fixed effects—is at most as strong as selection on the observed demographic controls. For the bound on the R^2 that would be obtained if both observable and unobservable determinants of the outcome were included as controls, we follow Oster’s empirical recommendation and set $R_{\max} = 1.3\tilde{R}$, where $\tilde{R}$ is the R^2 from the regression with the full set of observable controls. We report sensitivity across alternative $R_{\max}$ values in the appendix.

4.2 Results

Figure 1 displays the raw rate of exit from formal employment of outsourced and direct-hire workers. Figure 3 shows the differences in hazard rate between outsourced and direct-hire workers with controls successively added. We first add suboccupation fixed effects, then microregion-suboccupation-year fixed effects, then observable worker demographic characteristics such as gender, age, race, and education at spell start, and finally unobservable worker ability using AKM worker effects.⁶ Figure 4 shows the bias-adjusted estimates using the method pioneered by [Oster \(2019\)](#).

Cleaners. As shown in Figure 1, the hazard from formal employment of direct-hire cleaners is significantly higher than that of outsourced cleaners in almost all months during the first 4 years of employment, except for some unusual patterns in the first six months. For example,

⁶See Appendix A for details.

at the one-year mark, the raw 30-day hazard rate for outsourced cleaners is 3.7 percent, while it is 4.2 percent for direct-hire cleaners. The unusual pattern in the first three months is likely attributable to tenure-dependent employment protection legislation, which caused hazards to be significantly higher prior to the three-month mark than thereafter ([Arnold and Bernstein 2021](#)).⁷

The hazard rates of outsourced and direct-hire workers become closer over the course of the employment spell. They eventually cross at around four years of tenure, with the hazard of direct-hire workers becoming lower than that of outsourced cleaners thereafter.⁸ The difference in survival gradually attenuates thereafter, so outsourcing generally has a small but positive effect on employment survival probability (see Appendix Figure C.5).

Figure 3 shows that the estimates are broadly similar after successive addition of controls. This suggests that selection due to observable worker characteristics and labor market conditions do not drive these differences in hazard rates. Figure 4 shows that bias-adjusted estimates computed using method of [Oster \(2019\)](#) are highly similar.

Security guards. The effect of outsourcing on hazard from formal employment is larger and more negative for security guards. Figure 1 shows that with the exception of the first few months, outsourced security guards have much lower probabilities of transitioning from formal employment than direct-hire workers. At the one-year mark, the raw 30-day hazard rate for outsourced workers is 3.7 percent, while it is 1.8 percent for direct-hire workers. The hazard rates of outsourced and direct-hire workers become closer over the course of the employment spell, eventually narrowing to a statistically indistinguishable difference in the sixth year of tenure. The hazard estimates imply that outsourcing significantly increased the survival of employment spells for security guards (Appendix Figure C.5).

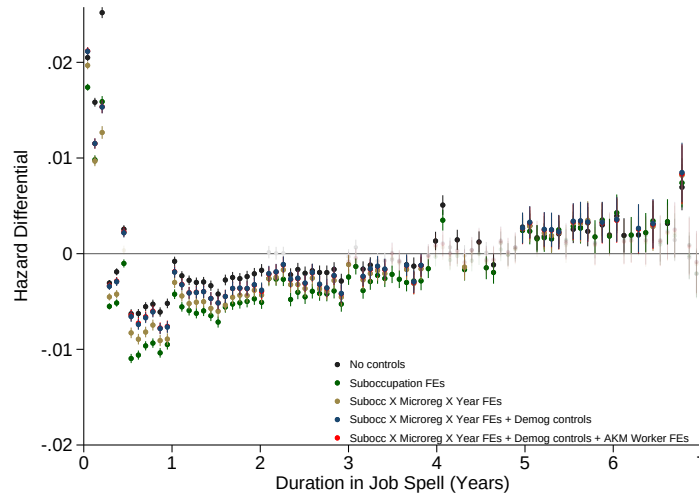
Figure 3 shows that adding suboccupation fixed effects attenuates the estimates somewhat, suggesting that for security guards, it is important to account for suboccupation differences.

⁷Specifically, the employer must pay a firing penalty in the event of an involuntary separation, which is equal to roughly one month of the worker's salary for every year the worker has been employed at the firm. The bulk of this penalty is paid to the worker as severance. This requirement is tenure-dependent and only applies after 3 months of employment.

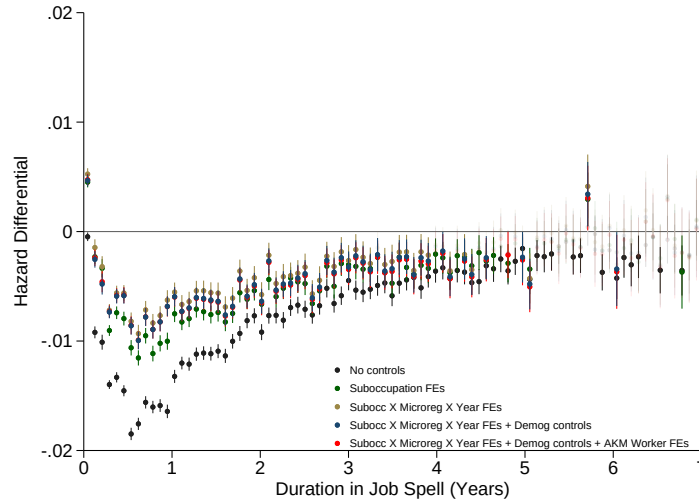
⁸As shown in Appendix Figure C.2, locally smoothed estimates of the outsourcing differential confirm that the hazard rate of outsourced cleaners "crosses" from above to below that of direct-hire cleaners at around four years of tenure.

Figure 3: Effect of Outsourcing on Hazard from Formal Employment, Alternative Controls

(a) Cleaners



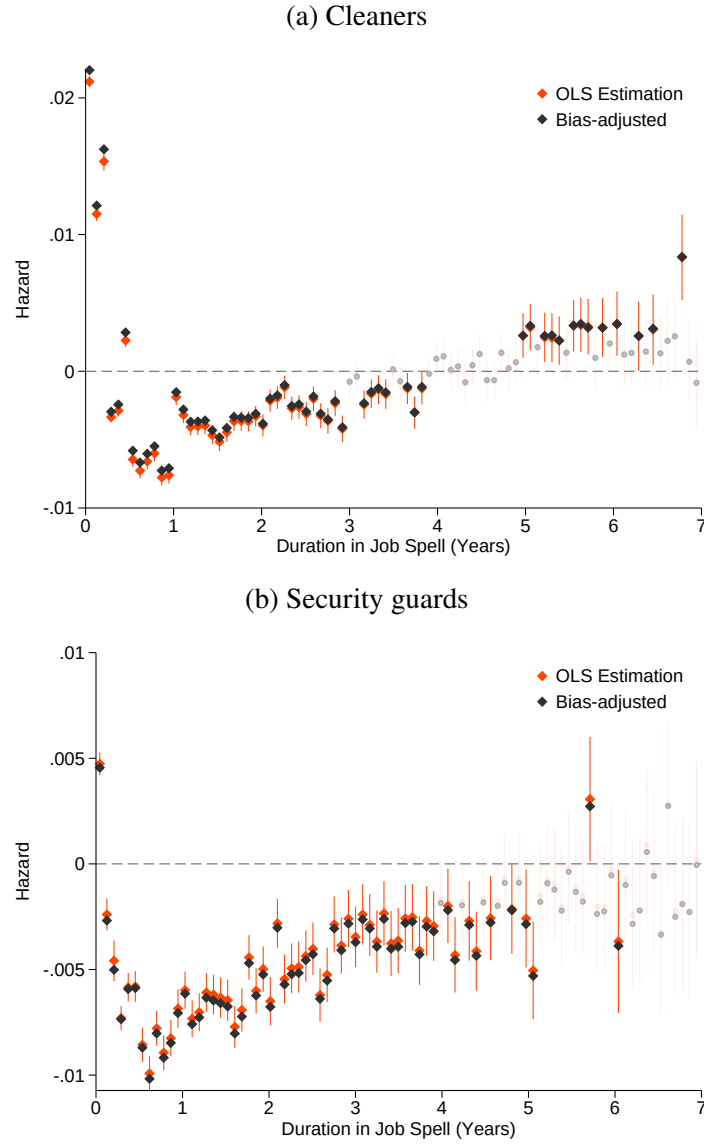
(b) Security guards



Notes: Sample includes all first full-time spells at each employer between 2003-2010. We truncate the duration at 7 years. Each hazard differential is estimated at the midpoint of the 30-day interval. The black dots display the raw difference in hazards. The green dots show estimates with only suboccupation fixed effects. The mustard dots show estimates with suboccupation X microregion X year fixed effects. The red shows the estimates with the full set of controls in our main regression. The blue dots show the estimates with the full set of controls and AKM worker effects estimated from a wage regression with the full sample. 95% confidence intervals are shown. Statistically insignificant estimates are shown in light gray.

However, the effects of outsourcing on hazard from formal employment among security guards remain large even after controlling for observable worker characteristics and labor market conditions. Figure 4 shows that estimates are highly similar after accounting for unobservable

Figure 4: Bias-adjusted Effect of Outsourcing on Hazard from Formal Employment



Notes: Following [Oster \(2019\)](#), we plot bias-adjusted estimates assuming that $\delta = 1$ and $R_{\max} = 1.3\tilde{R}$, after partialling out microregion-suboccupation-year fixed effects. The black dots display the bias-adjusted estimates. The orange dots show estimates with the full set of controls in our main regression and AKM worker effects estimated from a wage regression with full sample. 95% confidence intervals are shown for the OLS estimates. Statistically insignificant estimates are shown in light gray.

worker selection using the method of [Oster \(2019\)](#).

Robustness Checks. Appendix Table [C.2](#) shows that adding observable worker controls hardly changes the estimated coefficients even though it increases the R^2 , confirming that unobserved

worker selection is likely to be small. Appendix Figure C.1 shows that the bias-adjusted coefficients are nearly indistinguishable from OLS estimates across a wide range of $R_{\max}$ values.

Estimates with more restrictive samples or a less stringent definition of transitions from formal employment are reported in the appendix. Appendix Figure C.3 restricts to employment spells for which the individual initially entered from outside the formal sector, to workers who are no more than 30 at the start of the spell, and to male workers, respectively. Appendix Figure C.4 reports results where transitions from formal employment due to quits are included in the hazard definition.⁹ Appendix Figure C.5 shows the implied survival rates. Appendix Figure C.6 confirms that the results are robust to using the narrowest possible definition of outsourcing, restricting to a single CNAE code per occupation.

Our main hazard results define exits based on involuntary separations. One concern is that some separations labeled as involuntary may in fact be strategically relabeled quits. Using CAGED data (2017–2025), we find that mutual agreement separations are only 0.2–1.0 percentage points more common among outsourced workers, suggesting that relabeling is unlikely to substantially bias our estimates (see Appendix C.2).

4.3 Labor Demand Volatility and Employment Security

What explains the observed effects of outsourcing on the rate of exit from formal employment? Firm-level surveys show that firms outsource partly to overcome fluctuations in labor demand (Abraham and Taylor 1996; Houseman 2001).

In this subsection, we provide further evidence that outsourcing smooths labor market volatility. We measure city-level labor demand volatility, LDV_m , as the average of the absolute value of monthly change in city-level unemployment rate between 2003 and 2010. These measures are constructed from the PME, the Brazilian Monthly Employment Survey. These data are collected at a monthly frequency in six large metropolitan areas, enabling a less biased measurement of local labor demand volatility than is possible with the annually collected RAIS.¹⁰

⁹Relatedly, we find that outsourced workers are more likely than direct employees to experience employer-to-employer transitions (see Appendix Figure C.9). This could be because outsourced workers receive lower wages and therefore are more likely to accept an outside offer.

¹⁰The six metropolitan areas are Belo Horizonte, Porto Alegre, Recife, Rio de Janeiro, Salvador, and Sao Paulo.

We estimate the following regression:

$$y_{i\tau} = \alpha_\tau + \delta_\tau O_i + \gamma_\tau HighLDV_m + \chi_\tau HighLDV_m \times O_i + X_i' \beta_t + \theta_{ot} + \epsilon_{i\tau} \mid \tau_i \geq \tau, \quad (3)$$

where $HighLDV_m$ is a dummy indicating whether city m is among the three cities with higher labor demand volatility. Under the assumption that O_i and $HighLDV_m$ are conditionally independent of $\epsilon_{i\tau}$, the coefficients γ_τ capture the effects of living in a high-volatility city on employment hazard, while the coefficients χ_τ capture the additional effects of outsourcing on employment hazard in a high-volatility city. Our sample includes all first full-time spells at each employer between 2003-2010 in the relevant cities.

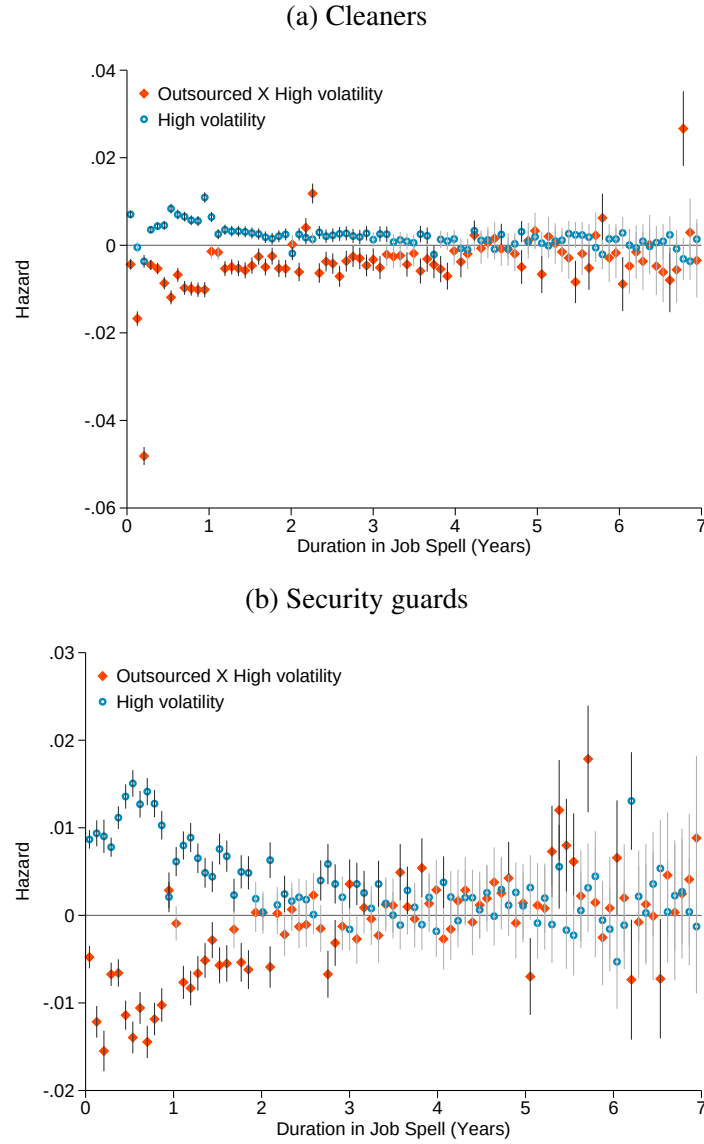
Figure 5 plots the estimated γ_τ and χ_τ coefficients. We find that employment hazards for direct-hire workers were substantially higher in high-volatility cities than in low-volatility cities. For cleaners, the increases are statistically significant during almost all months in the first three years. For the security guards, the increases are statistically significant during the first two years. The differences in hazard are especially large during the first year of employment tenure.

We also find that outsourcing reduces employment hazard more in cities with high volatility. Remarkably, the previously estimated increase in employment hazard in high-volatility cities is completely offset by outsourcing. As shown in Figure 5, the estimated γ_τ coefficients are the mirror image of the estimated χ_τ coefficients, suggesting that the positive effects of outsourcing on employment security are larger in more volatile labor markets. Appendix Figures C.7 and C.8 show that these results are robust to using a continuous measure of labor demand volatility as well as additional controls.

Why does outsourcing redress labor market volatility? One possibility is that outsourcing reduces employment hazard by enabling flexible reassignment of workers across firms. Consistent with this idea, Battiston, Espinosa and Liu (2021) shows that 2-4 percent of workers in a large Colombian security service firm are rotated across clients in each month. The next section builds a model based on this possibility.

Another possible mechanism is that professional service firms facilitate downward adjustment in wages. In other words, outsourced workers may be more likely than direct hires to

Figure 5: Effect of Outsourcing on Employment Hazard, High vs Low Volatility Labor Markets



Notes: Sample includes all first full-time spells at each employer between 2003-2010 in six major cities in Brazil: Belo Horizonte, Porto Alegre, Recife, Rio de Janeiro, Salvador, and Sao Paulo. Cities with higher labor market volatility than the median are classified as having high labor market volatility. The blue dot shows the estimated effect of working in cities with high labor market volatility on hazard to unemployment. The orange dot shows the added effect of working as an outsourced worker in cities with high labor market volatility on hazard to unemployment. We include the suboccupation X year fixed effects and baseline demographic controls. 95% confidence intervals are shown for the OLS estimates. Statistically insignificant estimates are shown in light gray.

experience wage cuts rather than layoffs in response to reductions in labor demand. Appendix Table B.2 rules out this alternative explanation. We regress log wage on local unemployment rate and its interaction with outsourcing status, controlling for worker demographics, as well

as suboccupation times microregion, and firm fixed effects. We find that the wages of outsourced workers are, if anything, more rigid than that of direct employees. Without controls for worker fixed effects, the negative relationship between wages and unemployment is similar for outsourced and direct-hire workers. With controls for worker fixed effects, the negative relationship between wages and unemployment becomes weaker for outsourced workers than for direct employees. This result suggests that outsourced contracts have greater wage rigidity. This may reflect the more standardized personnel policies of large organizations with many similar workers, or—consistent with the interpretive framework in Section 5—the possibility that intermediaries respond to client-specific demand shocks by reassigning workers across clients rather than by adjusting wages within a match.

A third possibility is that outsourced workers have higher match productivity than direct-hire workers, for example because intermediaries select workers positively, provide specialized training, or mitigate moral hazard problems. Higher initial match productivity would raise the continuation value of the match relative to the outside option and therefore lower the rate of exit from formal employment for outsourced workers. We cannot rule out such a productivity differential with our data, since we do not observe match productivity directly. However, Section 5 provides an interpretive framework in which the observed “crossing” pattern in hazard rates is consistent with reassignment and inconsistent with a level shift in match productivity.

5 Interpretive Framework

In this section, we shed further light on the underlying economic mechanisms by providing a stylized partial equilibrium search-and-matching interpretive framework. The framework offers two potential explanations for the observed lower hazard rates among outsourced workers. First, outsourcing alters the search and bargaining process in the labor market, enabling flexible reassignment of workers across clients, and consequently affecting workers’ wages and separation rates. Second, outsourced workers may differ in productivity compared to directly hired workers due to the selection processes of firms and workers into outsourcing, or due to training provided by intermediaries. Our main theoretical result is that hazard rates exhibit a “crossing” pattern

only when outsourcing eases worker reassignment. This result indicates that the observed hazard profiles align with the first explanation and not purely with the second.

5.1 Setup

Our model builds on [Blanchard and Landier \(2002\)](#), who study the effects of fixed-term contracts. The way we model reassignment of workers across firms by an intermediary is similar to models of on-the-job search ([Pissarides 1994](#); [Cahuc, Postel-Vinay and Robin 2006](#)).¹¹

Time is continuous with a discount rate r . A worker is matched with a firm under either employment or outsourcing. Under employment, the firm directly employs the worker. Under outsourcing, an intermediary employs the worker, but the worker is assigned to the firm. For simplicity, the choice between employment and outsourcing is assumed to be exogenous to the firm.¹² Under each arrangement $a \in \{E, O\}$, match productivity is initially y_a at $t = 0$. During the match, a single stochastic productivity shock z arrives at Poisson rate λ . The match productivity then changes to $y_a + z$, where z is a random variable with a continuous cumulative distribution $G(z)$.¹³

Wage is determined by Nash bargaining with worker bargaining parameter β_a through continuous renegotiation between the firm and the worker. Since wages are bargained, we say that match-specific rents are *shared* between the worker and firm. If bargaining fails, the worker receives an outside option $\bar{W}$. We make a simplifying assumption that intermediaries charge service fees that reduce the effective output of outsourced workers but have zero bargaining power. This assumption is plausible since intermediaries bid for service contracts competitively. Under this simplification, we abstract from the intermediary's problem and treat wages as determined by Nash bargaining between the client firm and the outsourced worker. The assumption should therefore be understood as a modeling simplification rather than a literal description of

¹¹Relatedly, [Shimer \(1999\)](#) and [Prat \(2006\)](#) study unemployment, worker turnover, and wage dispersion using models with match productivity shocks following Brownian motion. [Arnold and Bernstein \(2021\)](#) and [Cahuc, Malherbet and Prat \(2019\)](#) study the effects of discontinuities in severance pay schedules on worker hazard.

¹²Since we do not observe the identity of the end firm in our data, it is difficult for us to empirically analyze each firm's choice between outsourcing and employment.

¹³The change in productivity can be interpreted as symmetric learning about match productivity, as in [Jovanovic \(1979\)](#).

wage-setting.

Outsourcing has three potential effects in our model. First, outsourcing may alter worker bargaining power β_a . By allowing bargaining power to differ, we incorporate the prevailing notion in extant literature (e.g., [Dube and Kaplan 2010](#) and [Goldschmidt and Schmieder 2017](#)) that outsourcing lowers the rents that workers receive by circumventing within-firm fairness norms, avoiding collective bargaining agreements, or reducing efficiency wages.

Second, outsourcing allows workers to be reassigned across firms, while directly employed workers cannot. Specifically, we assume that if a worker-firm pair under outsourcing separates, then with some probability γ_O , the intermediary immediately matches the firm with a new worker and reassigns the worker to another firm. The productivity of the new match is y_O and another productivity shock z may arrive at rate λ . However, if worker-firm pairs under direct employment separate, then the worker receives her outside option and the firm must open a vacancy. In other words, the probability of reassignment under outsourcing is $\gamma_O \in [0, 1]$, while under employment $\gamma_E = 0$. For simplicity, we define $\gamma \equiv \gamma_O$.

Third, outsourcing may alter the match productivity level y_a . This could be because the firms that select into outsourcing are systematically different. For example, [Dube and Kaplan \(2010\)](#) and [Goldschmidt and Schmieder \(2017\)](#) document that high-wage firms are more likely to outsource. It could also be that outsourced workers are positively selected due to the hiring and matching expertise of the intermediary. In addition, intermediaries may provide additional training that enhances the labor productivity of outsourced workers.

5.2 Equilibrium under Employment and Outsourcing

Figure [D.10](#) visualizes the state transitions under arrangement a . At the initial state, workers' match productivity is y_a . Productivity shock z arrives at rate λ . There is a cutoff $\hat{z}_a$ such that separation will occur when z is below the cutoff. If $z \geq \hat{z}_a$, the match continues and no further productivity shock arrives. If $z < \hat{z}_a$, then with probability γ_a , the worker is reassigned to another firm with initial match productivity y_a , and another productivity shock z arrives at rate λ . Otherwise, the worker exits employment.

Let $\bar{V}^a$ denote the value of the firm that opens a vacancy under arrangement a . Let V_0^a denote the value of the firm that is matched with a new worker under arrangement a . Let $V_1^a(z)$ denote the value of a firm that remains matched with the worker after the productivity shock z arrives under arrangement a .

Once matched with a worker, the firm's Bellman equation before the shock is:

$$\begin{aligned}
rV_0^a = & \underbrace{(y_a - w_0^a)}_{\text{flow profit}} + \underbrace{\lambda \int_{\hat{z}_a}^{\infty} [V_1^a(z) - V_0^a] dG(z)}_{\text{stay after shock}} \\
& + \underbrace{(1 - \gamma_a)\lambda G(\hat{z}_a)(\bar{V}^a - V_0^a)}_{\text{separate after shock}} + \underbrace{\gamma_a\lambda G(\hat{z}_a)(V_0^a - V_0^a)}_{\text{re-assign after shock}}. \tag{4}
\end{aligned}$$

The flow profit for firms is productivity minus wages $(y_a - w_0^a)$. With probability λ , the productivity shock arrives. When $z > \hat{z}_a$, workers will stay after the shock, and firms' utility gain is $V_1^a(z) - V_0^a$. When $z < \hat{z}_a$, with probability $1 - \gamma_a$, firms and workers are separated and firms do not get reassigned a new worker. In this case, the utility loss is $\bar{V}^a - V_0^a$. With probability γ_a , firms get a new worker and the utility change is $V_0^a - V_0^a = 0$.

The worker's Bellman equation before the shock is:

$$\begin{aligned}
rW_0^a = & \underbrace{w_0^a}_{\text{flow wage}} + \underbrace{\lambda \int_{\hat{z}_a}^{\infty} [W_1^a(z) - W_0^a] dG(z)}_{\text{stay after shock}} \\
& + \underbrace{(1 - \gamma_a)\lambda G(\hat{z}_a)(\bar{W} - W_0^a)}_{\text{separate after shock}} + \underbrace{\gamma_a\lambda G(\hat{z}_a)(W_0^a - W_0^a)}_{\text{re-assign after shock}} \tag{5}
\end{aligned}$$

The flow payoff for workers is their wages. With probability λ , the productivity shock arrives. When $z > \hat{z}_a$, workers stay after the shock, and their utility gain is $W_1^a(z) - W_0^a$. When $z < \hat{z}_a$, with probability $1 - \gamma_a$, workers separate from the firm and take the outside option, and the utility loss is $\bar{W} - W_0^a$. With probability γ_a , workers are re-assigned to a new firm and their utility change is $W_0^a - W_0^a = 0$.

If the productivity shock arrives and there is no separation, the firm's Bellman equation after

the shock is

$$rV_1^a(z) = y_a + z - w_1^a(z) \quad (6)$$

The firm's flow profit is the post-shock productivity ($y_a + z$) minus the post-shock wage ($w_1^a(z)$).

If the worker remains matched with the same firm, the worker's Bellman equation is given by:

$$rW_1^a(z) = w_1^a(z). \quad (7)$$

Workers' flow utility is simply their wages. This is a self-absorbing state as no further productivity shocks will occur.

Wages are continuously negotiated through Nash bargaining, so we have that

$$(1 - \beta_a)(W_0^a - \bar{W}) = \beta_a(V_0^a - \bar{V}^a) \quad (8)$$

$$(1 - \beta_a)(W_1^a(z) - \bar{W}) = \beta_a(V_1^a(z) - \bar{V}^a) \quad (9)$$

The free-entry condition suggests that the value of firms opening a vacancy is always zero, i.e., $\bar{V}^a = 0$.

The matched worker-firm pair is indifferent between separation and continuation at the productivity cutoff $\hat{z}_a$, so the cutoff $\hat{z}_a$ is pinned down by $V_1^a(\hat{z}_a) = \bar{V}^a$. Hence, we have

$$\hat{z}_a = -y_a + r\bar{W} \quad (10)$$

The cumulative probability of endogenous separation is

$$F_a(t) = \left[1 - e^{-(1-\gamma_a)\lambda t} \right] G(\hat{z}_a) \quad (11)$$

The hazard rate (the probability of separation in the current period conditional on being with the firm in the last period) is:

$$h_a(t) = \frac{F_a'(t)}{1 - F_a(t)} = \frac{(1 - \gamma_a)\lambda G(\hat{z}_a)}{G(\hat{z}_a) + (1 - G(\hat{z}_a))e^{(1-\gamma_a)\lambda t}} \quad (12)$$

Workers' wages before and after the match productivity shock arrives are, respectively:

$$w_0^a = \beta_a y_a + (1 - \beta_a) r \bar{W}, \quad (13)$$

and

$$w_1^a(z) = \beta_a (y_a + z) + (1 - \beta_a) r \bar{W}. \quad (14)$$

5.3 Theoretical Predictions

In Section 4, we documented a striking “crossing” pattern in the hazard among cleaners in Brazil, wherein the hazard from formal employment of outsourced cleaners is initially lower than similar direct-hire cleaners but becomes higher after the first few years of employment tenure. Proposition 1 shows that this “crossing” pattern can be rationalized in our model by the possibility of flexible reassignment of workers across firms.

Proposition 1. *If the probability of reassignment under outsourcing, γ_O , is sufficiently large, then there exists some T such that $h_E(t) \geq h_O(t)$ if and only if $t \leq T$. Otherwise, $h_E(t) \leq h_O(t)$ for all t .*

Proof. See appendix. □

The intuition for Proposition 1 is as follows. Combining Equations (10) and (12), we can show that the initial hazard rate $h_a(0) = (1 - \gamma_a) \lambda G(-y_a + r \bar{W})$. It follows that $h_O(0) < h_E(0)$ if either γ_O is large or y_O is high relative to y_E . In other words, transitions from formal employment are less likely at the start of an employment spell under outsourcing either if the intermediary reassigns workers across clients or if the initial worker-firm match productivity is larger for outsourced workers than direct-hire workers.

Recall that the reassignment probability for direct hires is zero, $\gamma_E = 0$. If the probability of reassignment under outsourcing (γ_O) is large, however, the slope of the hazard respective to time under outsourcing flattens. This is because reassignment would lead to a new match at the initial match productivity level, and another match-specific shock may yet arrive. As a result,

there is a form of dynamic selection: Conditional on survival, outsourced workers are less well protected from match-specific shocks. This causes their hazard rates to fall more slowly than those of directly employed workers.¹⁴

The combination of these two effects explains why there exists a cutoff T such that $h_E(t) \geq h_O(t)$ if and only if $t \leq T$ only if γ_O is sufficiently large. If the cutoff T exists, it is possible to derive an explicit formula for it:

$$T = \frac{1}{\lambda} \log \left[\frac{\gamma_O}{e^{1-\gamma_O}(1 - 1/G(\hat{z}_O)) - (1 - \gamma_O)(1 - 1/G(\hat{z}_E))} \right] \quad (15)$$

An immediate implication is that T increases in y_O , but decreases in y_E and λ . The relationship between T and γ_O is more ambiguous. It can be shown that T increases in γ_O if and only if $e^{1-\gamma_O}(1 + \gamma_O) > \frac{G(\hat{z}_O)}{1-G(\hat{z}_O)} \cdot \frac{1-G(\hat{z}_E)}{G(\hat{z}_E)}$. Therefore, T increases in γ_O if $y_O \leq y_E$. However, T may be decreasing in γ_O if y_O is much larger than y_E .

In summary, our model predicts that two channels could drive the lower initial hazard rates for outsourced workers: 1) $\gamma_O > 0$, and 2) $y_O > y_E$. However, if outsourcing solely impacts productivity ($y_O > y_E$) without influencing reassignment ($\gamma_O = 0$), the crossing pattern observed in the hazard rates would not occur. The observation that the hazard rate for outsourced cleaners is initially lower and subsequently becomes higher than that for direct-hired cleaners suggests that reassignment plays a significant role. While this does not exclude the productivity channel, it indicates that the channel of higher initial match productivity is insufficient to explain the hazard patterns observed in the data.

6 Net Present Values for Outsourced and Direct-hire Workers

To assess the significance of the differing hazard rates between outsourced and directly hired workers, we compute the net present value (NPV) for these two groups. This analysis considers

¹⁴The assumption of a one-time shock is not important for this result. As long as the probability of a subsequent negative match-specific shock falls after a match survives an initial shock, the hazard rates of the outsourced worker will fall more slowly due to the possibility of reassignment. It is only if the probability of a negative shock does not diminish conditional on survival (e.g., match productivity is drawn i.i.d. over time) that this prediction no longer applies.

not only their wage differentials but also the differences in hazard rates, which influence their NPV over the entire life cycle.

6.1 NPV with the E-U Transition

We first start with a simple model where the only possible transition is the employment-to-unemployment (E-U) transition. We will consider the employment-to-employment (E-E) transition in the next section.

Let $h^a(t)$ represent the E-U hazard rate, and $w^a(t)$ denote the wage profile of a worker of type a at tenure t . From RAIS, we obtain the hazard rates and wage profiles by tenure. Figure 3 presents the E-U hazard rates by tenure, while Figure B.2 shows the wage profiles by tenure.

Based on the hazard rate $h^a(t)$, we derive the survival rate $s^a(t)$, which captures the probability that a worker of type a remains with their current employer at tenure t :

$$s^a(t) = s^a(t-1)(1 - h^a(t)) \quad (16)$$

The NPV of a worker of type $a \in \{OC, EC, OG, EG\}$ (corresponding to outsourced cleaners, direct-hired cleaners, outsourced guards, and direct-hired guards) at the start of employment is given by

$$NPV^a = \sum_{t=0}^T \beta^t (s^a(t)w^a(t) + (1 - s^a(t))\tilde{w}_{EU}^a(t, d)) \quad (17)$$

where $\tilde{w}_{EU}^a(t, d)$ is the wage received by a worker after the E-U transition, and d denotes the period when the worker separates. In our calculation, each period is one month, and T is set to be 72 months.¹⁵ We set the monthly discount rate $\beta = 0.9975$.

The post-separation wage is defined as

$$\tilde{w}_{EU}^a(t, d) = w^a(d) * (1 - \delta_{EU}^a(t - d)) \quad (18)$$

where $w(d)$ is the wage at the time of separation, and $\delta_{EU}^a(t - d)$ represents the wage penalty

¹⁵In our sample, the average tenure of guards and cleaners is 4 years, and 75% of workers have a tenure below 6 years.

from the E-U transition, which depends on the duration of the separation $\tau = t - d$.

Wage Imputation Since we can only observe wages for formal workers in RAIS, non-employed workers in RAIS may also include some individuals employed in the informal sector. To account for informality, we also impute wages for workers not formally employed. First, we estimate the share of non-formally-employed workers engaged in the informal sector using data from PNADc (Pesquisa Nacional por Amostra de Domicílios Contínua). The PNADc is a quarterly panel survey available from 2012 to 2019 that covers workers in the informal sector.¹⁶ Using PNADc, we estimate the transition rates from formal outsourced or directly hired positions to informal employment, denoted as $P_{a \rightarrow informal}$, which varies with worker type a . Appendix Table E.6 reports the results for $P_{a \rightarrow informal}$. Approximately 60% to 70% of workers switch from formal employment to informal employment after a layoff. This ratio is 3–4 percentage points higher for direct-hired workers compared to outsourced workers, and about 5 percentage points higher for guards than for cleaners.

Second, we estimate the wage penalty of working in the informal sector compared to the formal sector, denoted as γ_a , using data from PNADc. Appendix Table E.7 shows that the wage gap between informal and formal employment remains relatively stable across different job types, with informal workers earning roughly 12% less than their formal counterparts.

The imputed wage of workers not observed in formal employment is given by:

$$\hat{w}_{it} = P_{a \rightarrow informal} \times (w_{i,\tau} \times e^{\gamma_a}) + (1 - P_{a \rightarrow informal}) \times 0 \quad (19)$$

where $w_{i,\tau}$ is the last observed wage at time τ , $P_{a \rightarrow informal}$ is the probability of transitioning to informal employment for worker type a , and γ_a is the corresponding informal-formal log wage differential. Appendix E provides more details regarding the computation of the imputed wage.

Estimating the E-U Wage Effect After imputation, we estimate wage loss from the E-U transition, $\delta_{EU}^a(\tau)$, through an event study analysis. Our sample includes workers who experienced

¹⁶Although this period does not overlap with RAIS (which covers 2003–2010), PNADc is the only dataset with a panel structure with information on informal employment.

an E-U transition, and we match them with a control group comprising workers who either experience their first event in a later year (not-yet-treated) or do not experience the event at all during the sample period (never-treated). Importantly, we ensure that control workers are employed in the same occupation as the treated workers at $t = -1$.

For the matching process, we first require exact matching on the worker’s employment history during the two years prior to the event (at times $t = -1$ and -2). Next, we select a control worker with the closest match to the treated worker based on age, years of schooling, and wages at $t = -1$ and -2 . Using a nearest neighbor approach, we assign one control worker to each treated worker. This combination of distance-based and exact matching ensures that treated and control workers are highly comparable in terms of their observable characteristics and recent labor market trajectories.

We estimate the following specification using RAIS data:

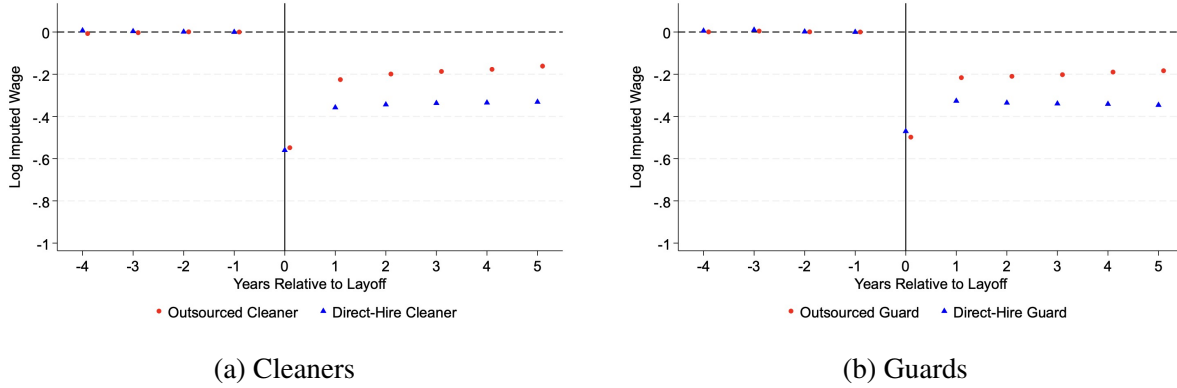
$$\hat{w}_{it}^a = \sum_{\tau \neq -1} \delta_{EU}^a(\tau) \cdot D_{it}^\tau \cdot T_i + \sum_{\tau \neq -1} \theta_\tau \cdot D_{it}^\tau + \xi \cdot T_i + \alpha_i + \lambda_t + \varepsilon_{it} \quad (20)$$

where T_i is an indicator for belonging to the treatment group; D_{it}^τ is an indicator for being τ years from the event (with $\tau = -1$ as the omitted reference period); α_i are worker fixed effects and λ_t are calendar year fixed effects. Standard errors are clustered at the worker level.

The results using imputed wages are presented in Figure 6. They reveal a substantial wage loss following layoffs. There is an immediate decrease in imputed wages right after involuntary separation—approximately 55% for outsourced and direct-hired cleaners, and 50% for outsourced and direct-hired guards. However, outsourced workers recover significantly faster than their direct-hired counterparts. Five years after the layoff, outsourced cleaners (guards) experience a 16% (18%) wage loss compared to their pre-layoff level, whereas direct-hired cleaners (guards) suffer from a 33% (35%) wage reduction. This suggests an additional advantage of outsourcing: it provides more opportunities for displaced workers to find better jobs and helps mitigate the long-term negative impact of layoffs on wages.

Appendix Figure E.11 presents the event study results of the E-U transition for original wages. To address the issue of zero wages for workers not formally employed, we use the wage

Figure 6: Wage Effects of E-U Transition (Imputed Wages)



ratio (current wage relative to the wage at $t = -1$) instead of the log wage as the outcome variable. Without imputing wages, all workers not in the formal sector are treated as unemployed, resulting in a much larger estimated wage loss from the E-U transition. Nonetheless, we still observe that outsourced workers recover much faster from layoffs compared to direct-hired workers.

NPV Results Table 3 presents the NPV results. The first row considers a scenario without E-U and E-E transitions, capturing only the wage differences between outsourced and direct-hired workers – this aligns with the approach in previous literature. We normalize the NPV of direct-hired workers to one and report only the relative NPV for outsourced workers. When focusing solely on wages and excluding turnover effects, outsourced workers have a lower NPV compared to direct-hired workers, as they earn lower wages. Specifically, the NPV of outsourced cleaners is 88.5% of that of direct-hired cleaners, and the NPV of outsourced guards is 96.3% of their direct-hired counterparts. The gap is larger for cleaners, reflecting their bigger wage disparity.

Table 3: NPV of Outsourced Workers Relative to Direct-hired

	Cleaners	Guards
No EU or EE hazard	0.885	0.963
EU hazard, imputed wage	0.940	1.017
EU hazard, original wage	0.983	1.068
EU and EE hazards, imputed wage	1.013	1.028
EU and EE hazards, original wage	1.077	1.074

The second row introduces the consideration of the E-U hazard. For guards, the NPV of outsourced workers becomes 1.7% higher than that of direct-hired guards. For cleaners, although the NPV remains lower for outsourced workers, the gap narrows from 88.5% to 94.0%, indicating a reduced disparity when accounting for the E-U transition hazard.

The third row presents a robustness check using the original wage instead of the imputed wage. The qualitative conclusion remains consistent, while the NPV of outsourced workers relative to direct-hired workers improves slightly further.

6.2 Adding the E-E Transition

In this section, we additionally consider the employment-to-employment (E-E) transition. In this case, the survival rate for workers of type a at period t is given by:

$$s^a(t) = s^a(t-1)(1 - h^a(t))(1 - o^a(t)) \quad (21)$$

where $o^a(t)$ is the probability of an E-E transition for a worker of type a in period t . Figure C.9 presents the E-E hazard rates for the four types of workers. Outsourced workers exhibit higher E-E hazards compared to direct-hired workers. This may be because their current wages are lower and/or they have a higher likelihood of receiving outside offers facilitated by intermediaries.

The shares of workers who have experienced E-U and E-E transitions by period t are denoted by:

$$s_{EU}^a(t) = s_{EU}^a(t-1) + s^a(t)h^a(t) \quad (22)$$

$$s_{EE}^a(t) = s_{EE}^a(t-1) + s^a(t)o^a(t) \quad (23)$$

The NPV of a worker is given by

$$NPV^a = \sum_{t=0}^T \beta^t (s^a(t)w^a(t) + s_{EU}^a(t)\tilde{w}_{EU}^a(t, d) + s_{EE}^a(t)\tilde{w}_{EE}^a(t, d)) \quad (24)$$

where $\tilde{w}_{EU}^a(t, d)$ and $\tilde{w}_{EE}^a(t, d)$ represent the wages of workers who have experienced E-U and

E–E transitions, respectively. d denotes the period when the worker experiences an E-U or E-E transition.

The post-EE wage is defined as

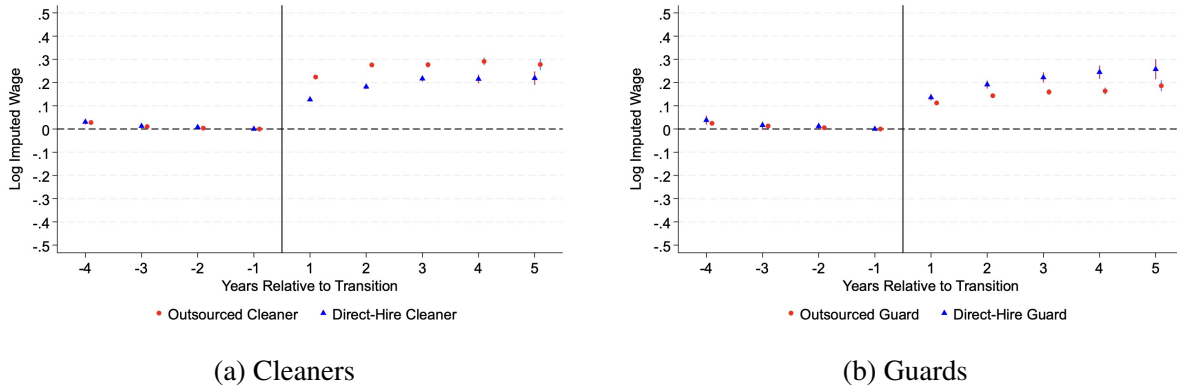
$$\tilde{w}_{EE}^a(t, d) = w^a(d) * (1 - \delta_{EE}^a(t - d)) \quad (25)$$

where $w(d)$ is the wage at the time of separation, and $\delta_{EE}^a(t - d)$ represents the wage penalty from the E-E transition, which depends on the duration of the separation $\tau = t - d$.

E-E Transition Rates Figure C.9 plots the difference in E-E transition rates between outsourced and direct-hired workers. For both cleaners and guards, outsourced workers exhibit higher E-E transition rates, with a difference of approximately 3 percentage points for cleaners and 0.4 percentage points for guards. Two possible explanations exist for this pattern. First, outsourced jobs may have lower value than direct-hire positions, prompting workers to quit more readily when they receive outside offers. Second, outsourced workers might be more likely to receive outside offers in the first place, as intermediaries could provide more job opportunities. Our NPV estimates will help distinguish between these two channels.

Estimating the E-E Wage Effect We follow the same matching approach to construct the control group and the same event study specification as in the E-U wage effect (Equation (20)) to estimate $\delta_{EE}^a(t - d)$, using RAIS data. The results using imputed wages are presented in Figure 7. For cleaners, the transition results in an average wage increase of 27% within the first five years for outsourced workers, which is higher than the 19% increase observed for direct-hired workers. Conversely, for guards, the pattern is reversed: direct-hired guards experience a higher wage increase of 21%, compared to 15% for outsourced guards. One possible explanation is that outsourced guards already earn wages closer to their outside option, leaving less room for improvement upon switching employers. Consistent with this, the outsourcing wage penalty is smaller for guards than for cleaners (Section 3). Figure E.12 presents the event study results using original wages, and the observed pattern remains consistent, indicating robustness of the findings.

Figure 7: Wage Effects of E-E Transition (Imputed Wages)



NPV Results The last two rows of Table 3 present NPV results incorporating E-U and E-E transitions, using imputed wages and original wages, respectively. Because outsourced workers have higher E-E hazard rates than their direct-hired counterparts, and because E-E transitions lead to wage growth, including these effects further increases the NPV for outsourced workers relative to direct-hired workers. Consequently, the NPV for outsourced workers is consistently higher: 1.3% higher for cleaners and 2.8% higher for guards relative to their direct-hired counterparts. Using original wages yields similar results, with the NPV gap being even more pronounced. These higher NPV estimates also suggest that outsourced workers' elevated E-E transition rates are likely driven by a greater chance of receiving outside offers through intermediaries.

These findings jointly suggest that it is important to consider not only the wage differential between outsourced and direct-hired workers but also their differences in hazard rates. Focusing solely on the wage gap would imply that the NPV of direct-hired workers is higher. However, when accounting for both E-U and E-E transitions, the NPV of outsourced workers surpasses that of direct-hired workers. This reversal is consistent with the reassignment channel emphasized in our interpretive framework: outsourcing firms may redeploy workers across client firms, reducing separation risk and enabling faster recovery from job loss. Models that treat outsourcing as purely a wage-reduction device would not capture this channel.

7 Conclusion

This paper presents the first estimates of the effects of domestic outsourcing on worker employment security. Using comprehensive administrative data on security guards and cleaners in Brazil, we first confirm that outsourcing is associated with lower wages, especially for low-wage workers, as suggested by recent literature (Dube and Kaplan 2010; Goldschmidt and Schmieder 2017; Drenik et al. 2023). We then robustly find that outsourcing is associated with a much lower rate of exit from formal employment during the first few years of employment spells. This difference is not explained by observable worker characteristics or differential exposure to labor market conditions. Accounting for both employment-to-unemployment and employment-to-employment transitions, we estimate that the net present value of earnings for outsourced cleaners and guards is approximately equal to or slightly higher than that of direct-hire workers—reversing the ordering implied by wage differentials alone.

Moreover, we find that these effects are large in labor markets with greater demand volatility. We also provide evidence that these effects are not due to outsourcing enabling flexible wage adjustments. These patterns are consistent with a simple search-theoretic model wherein intermediary firms reassign outsourced workers across client firms in the event of negative productivity shocks. In our interpretive framework, outsourced workers may differ from direct hires in match productivity—for instance, through selection, training, or reduced moral hazard—but such productivity differences alone cannot generate the observed “crossing” pattern in cleaner hazard rates; reassignment is a plausible mechanism that can.

Our findings are important since the existing literature on the aggregate effects of outsourcing ignores the potential gains to workers from increased employment security and flexible reassignment of workers across firms. Our findings suggest that outsourcing has less negative consequences for workers than implied by prior literature. Future studies on the effects of domestic outsourcing on worker earnings should account for this potential benefit of domestic outsourcing, especially for non-core activities where contract-firm economies of scale in reassignment can be large and in developing country contexts where turnover is high.

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Appendix

A Data Definitions

Outsourcing definition. Appendix Tables A.1 and A.2 show our classification of occupation and industry codes. Appendix Table A.3 shows that the outsourced share of security guards steadily grew from 48 percent to 70 percent between 1998 and 2016. By comparison, there was only modest growth in outsourced employment of cleaners during the same period, which grew from 34 percent to 37 percent.

Estimating AKM effects. To construct worker and firm wage components, we use the two-way decomposition method of Abowd, Kramarz and Margolis (1999) (henceforth, “AKM effects”). Using data on all formal workers in RAIS spanning 1998-2016, we estimate:

$$\log w_{it} = \psi_{J(i,t)} + \alpha_i + \theta_t + X_{it}\beta + \epsilon_{ijt},$$

where w_{it} represents real monthly wage, α_i is an individual fixed effect (capturing the general productive characteristics of workers), $\psi_{J(i,t)}$ is a firm fixed effect (capturing the wage premia for all workers at the firm), θ_t is a year fixed effect, $X_{it}\beta$ are the effects of time-varying observable worker characteristics (such as education and age), and ϵ_{ijt} is a composite error that may include idiosyncratic worker-firm match effects.

The estimated “AKM firm effect” ($\hat{\psi}_j$) can be thought of as representing the time-invariant pay premium of a given firm. The estimated “AKM worker effect” ($\hat{\alpha}_i$) can be thought of as representing time-invariant unobserved worker ability. To ensure that firm and worker fixed effects are identified, we restrict our analysis to the largest connected set of firms that are linked by workers moving between them.¹⁷ A further concern when estimating the AKM model is limited mobility bias, which may generate misleading variance decompositions, as discussed by

¹⁷Identification of the AKM model requires that workers do not move across firms in a manner that is systematically correlated with unmeasured productivity (Gibbons and Katz 1992). Alvarez et al. (2018) provide evidence that this assumption is justified in Brazilian RAIS data.

Andrews et al. (2008). We use a long panel so that limited mobility bias is more likely to be small (Bonhomme et al. 2023; Lachowska et al. 2023).

Table A.1: Occupation Classifications

Classification	CBO code	Description
Guard	517215	Municipal civil guard (<i>Guarda Civil Municipal</i>)
Guard	517310	Security agent (<i>Agente de Segurança</i>)
Guard	517330	Security guard (<i>Vigilante</i>)
Guard	517420	Watchperson (<i>Vigia</i>)
Cleaner	514210	Cleaner (<i>Faxineiro</i> ; deactivated Jan 2009)
Cleaner	514225	Cleaning and preservation worker, public areas (<i>Trabalhador de Serviços de Limpeza e Conservação de Áreas Públicas</i>)
Cleaner	514320	Cleaner (<i>Faxineiro</i> ; active from Jan 2009)

Notes: CBO (*Classificação Brasileira de Ocupações*) is the Brazilian Classification of Occupations established by the Ministry of Labor to identify occupations in the labor market.

Table A.2: Contract Firm Classifications

Classification	CNAE Code	Description
Contract firm	74160	Business management advisory activities
Contract firm	74500	Selection, agency and hire of labor
Contract firm	74608	Investigation, surveillance and security activities
Contract firm	74705	Activ. of hygiene and cleaning in buildings
Contract firm	74993	Other activ. of serv. provided mainly to other companies

Notes: CNAE, National Classification of Economic Activities, is the official industry classification used by statistics and by federal, state, and municipal bodies in Brazil.

Table A.3: Trend in Outsourcing as Measured by Contract-firm Employment

Contract-firm share of employment		
Year	Cleaners	Guards
1998	33.5%	48.0%
1999	33.3%	52.1%
2000	36.7%	53.5%
2001	31.2%	55.1%
2002	31.4%	57.2%
2003	33.4%	57.9%
2004	34.2%	58.0%
2005	35.0%	58.6%
2006	34.8%	59.5%
2007	34.7%	60.0%
2008	37.9%	61.0%
2009	37.6%	62.3%
2010	37.2%	63.7%
2011	37.4%	64.6%
2012	37.2%	66.2%
2013	37.6%	67.5%
2014	37.0%	67.6%
2015	36.2%	68.6%
2016	36.5%	69.8%
Change	3.1%	21.9%

Table A.4: Comparison of Formal and Informal Workers, Brazil, 2003–2010

(a) Cleaners

	Formally employed	Informally employed	Difference (with controls)
Age	38.03 (11.34)	35.66 (13.58)	-1.71** (0.21)
Years of schooling	6.06 (3.54)	5.44 (3.61)	-0.53** (0.06)
Male	0.46 (0.50)	0.44 (0.50)	-0.05** (0.01)
Non-white	0.59 (0.49)	0.66 (0.48)	0.01 (0.01)
Working hours	42.93 (8.72)	36.40 (14.00)	-6.77** (0.21)
Log wage	2.63 (0.36)	2.47 (0.57)	-0.09** (0.01)
Observations	33352	10737	

(b) Security guards

	Formally employed	Informally employed	Difference (with controls)
Age	38.44 (11.23)	39.82 (14.52)	1.61** (0.28)
Years of schooling	7.64 (3.53)	6.03 (3.95)	-1.41** (0.07)
Male	0.96 (0.19)	0.96 (0.21)	-0.01* (0.00)
Non-white	0.55 (0.50)	0.65 (0.48)	0.03** (0.01)
Working hours	47.28 (11.57)	46.94 (17.49)	-0.67* (0.32)
Log wage	2.90 (0.45)	2.54 (0.61)	-0.21** (0.01)
Observations	33892	7024	

Notes: This table shows the formally and informally employed comparison for cleaners and security guards. To define formally employed, the variable legally employed in the dataset is used. Occupational code for cleaners is 5142, while for security guards are 5173 and 5174. For controlled difference, year and state fixed effects are included. For the log wage, age, age square, years of schooling, male and non-white are included as controls, in addition to year and state dummies. Standard errors are displayed in parentheses, with ~ = significant at the 10% level, * = significant at the 5% level, and ** = significant at the 1% level. Source: PNAD dataset (Brazil's National Household Sample Survey).

Table A.5: Firm Characteristics by Occupation and Outsourcing Status, 2010

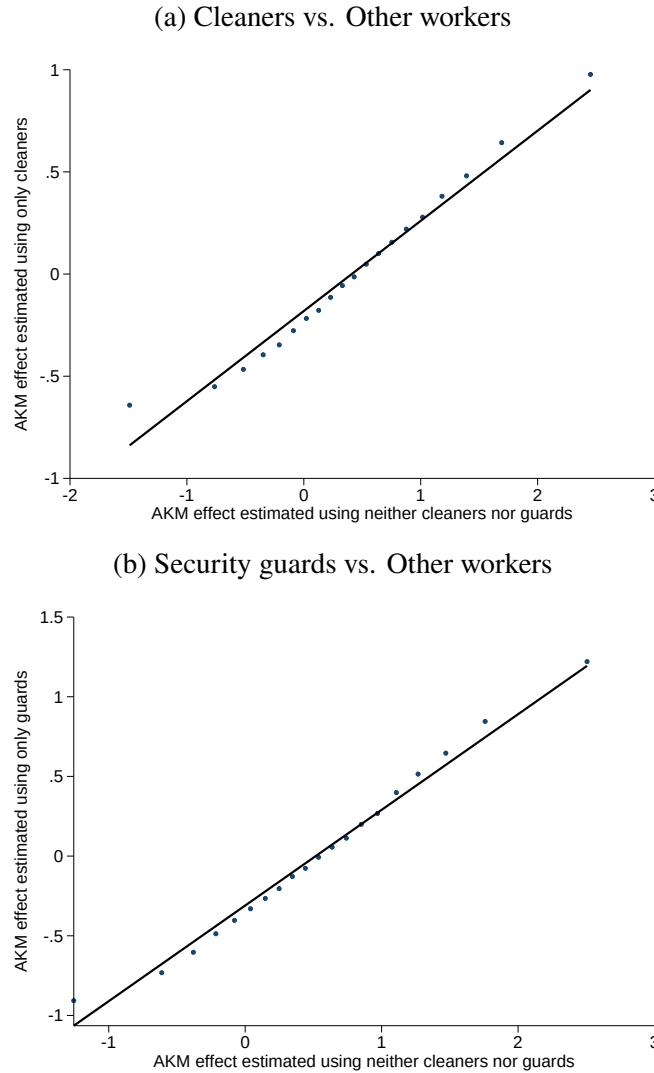
	Cleaners			Security Guards		
	Direct-hire	Outsourced	Ratio	Direct-hire	Outsourced	Ratio
<i>Panel A: Firm size (Dec 31 headcount)</i>						
Mean	3,318	3,472	1.05	9,392	3,067	0.33
Median	202	1,614	7.99	724	1,541	2.13
P25	17	414		77	601	
P75	1,214	4,387		3,348	3,372	
<i>Panel B: Firm age (years since first appearance in RAIS)</i>						
Mean	16.5	16.1	0.98	17.2	16.5	0.96
Median	19	15		23	17	
P25	8	10		9	10	
P75	25	25		25	25	
<i>Panel C: Same-occupation workers at firm</i>						
Mean	456	2,044	4.48	719	2,572	3.58
Median	31	919	29.6	48	1,379	28.7
P25	3	240		6	535	
P75	214	2,570		265	3,014	
<i>Panel D: Same-occupation share of firm headcount</i>						
Mean	0.28	0.63		0.21	0.90	
Workers	1,191,187	408,361		553,033	490,492	

Notes: Each worker contributes one observation with their firm's characteristics. Firm size is the number of unique workers employed at the firm on December 31, 2010 (all occupations). Firm age is 2010 minus the first year the firm appears in RAIS, scanning back to 1995; firms first observed in 1995 are right-censored (31–47% of workers, depending on group). Same-occupation workers counts the number of cleaners (for cleaners) or security guards (for guards) at the worker's firm on December 31, 2010. Same-occupation share is the ratio of same-occupation workers to total firm headcount. Data source: RAIS, 2010 cross-section.

B Additional Wage Results

B.1 Firm Wage Premia, Wage-Tenure Profile, and Wage Responsiveness

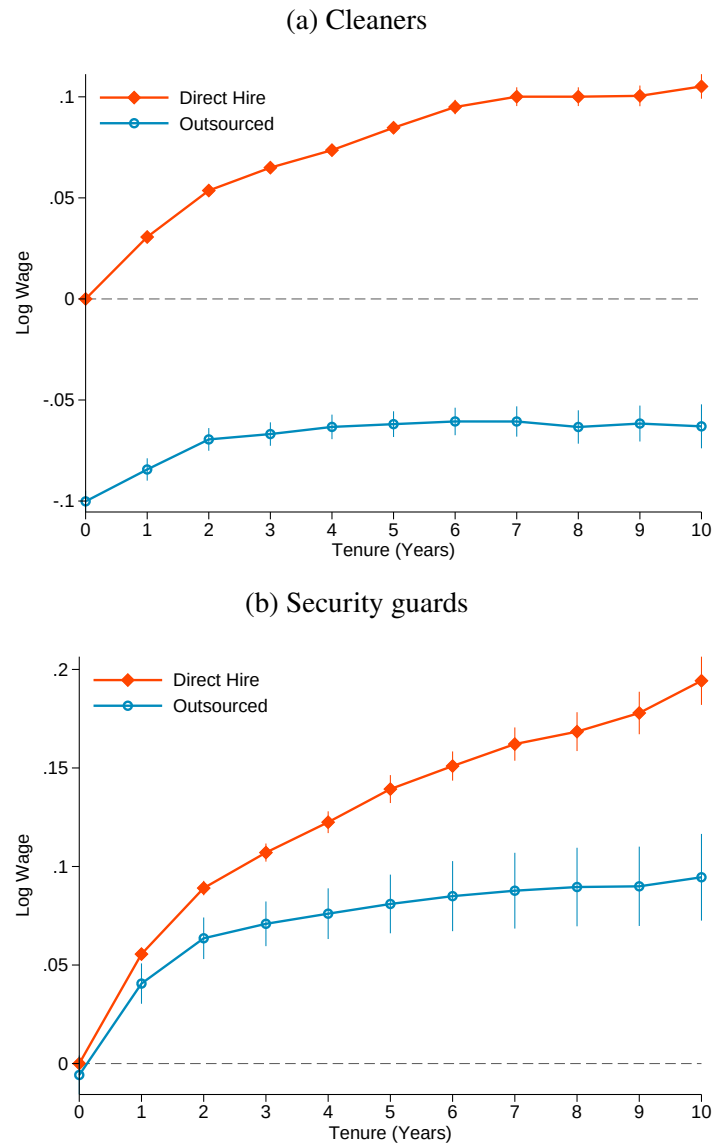
Figure B.1: Comparing Estimated Firm-level Wage Premia using Different Subsamples



Notes: The figure shows a binned scatter plot of standardized AKM firm effects estimated using cleaners and other workers neither cleaners nor security guards, and another plot of standardized AKM firm effects estimated using security guards and other workers neither cleaners nor security guards. Each dot corresponds to 1/20 of the observations. We run a simple regression first and plot the fitting line in this figure.

Following [Goldschmidt and Schmieder \(2017\)](#), we use a split sample IV approach to correct measurement errors in the RHS. For cleaners, the regression coefficient of the simple regression is 0.441 (SE 0.0017). And the coefficient of IV regression is 0.606 (SE 0.0026). For security guards, the regression coefficient of the simple regression is 0.600 (SE 0.0033). And the coefficient of IV regression is 0.970 (SE 0.0050). All standard errors clustered on the firm level.

Figure B.2: The Effects of Outsourcing on Wage-tenure Profile



Notes: This figure plots the estimated effects of outsourcing on wage-tenure profile. We plotted the estimated coefficients of the interaction terms between the tenure year dummies and the dummy indicating outsourcing in a wage tenure regression. We set the interaction term between the dummy indicating tenure less than 1 year and the dummy indicating workers are direct hired as zero to work as the reference group in the regression.

Table B.1: Wage Tenure Regression

	Cleaners		Security guards	
Outsourced	-0.110 (0.003)	-0.105 (0.003)	-0.013 (0.005)	0.005 (0.006)
Tenure		0.011 (0.000)		0.021 (0.001)
Outsourced X Tenure		-0.005 (0.001)		-0.012 (0.001)
Observations	6150003	6150003	4253501	4253501
R^2	0.94	0.94	0.93	0.93
Occ X Year X Microregion FE	X	X	X	X
Demographic controls	X	X	X	X
Worker FE	X	X	X	X

Notes: Dependent variable is log wage. Sample and controls are the same as Table 2 column (3). Tenure is measured in a continuous variable. Standard errors are displayed in parentheses, clustered at both worker and firm level.

Table B.2: Wage Regression, Controlling State Level Unemployment Rate

	Cleaners		Security guards	
State Level Unemployment Rate	-5.145 (0.093)	-0.980 (0.086)	-5.220 (0.252)	-1.614 (0.150)
Outsourced X State Level Unemployment Rate	0.757 (0.203)	1.250 (0.136)	2.201 (0.369)	2.130 (0.323)
Observations	7648116	6086694	4708677	4220416
R^2	0.72	0.94	0.79	0.94
Occ X Microregion FE	X	X	X	X
Demographic controls	X	X	X	X
Firm FE	X	X	X	X
Worker FE		X		X

Notes: Dependent variable is log wage. The sample is the same as Table 2 column (3). We control state-level annual unemployment rates between 2003 and 2010, which are calculated from PNAD data. We interact the unemployment rate with the outsourcing dummy. The unemployment rate in 2010 is imputed by taking the average unemployment rate between 2009 and 2011. Standard errors are displayed in parentheses, clustered at both worker and firm level.

B.2 Non-Wage Benefits

Following [Corbi, Ferraz and Narita \(2025\)](#), we use the annual PNAD surveys (2003–2010), which include information on non-wage benefits, to examine whether outsourced workers are less likely to receive employer-provided benefits. Our analysis focuses on five types of benefits: housing assistance, food assistance, transportation assistance, education or childcare assistance, and health or rehabilitation assistance. The benefit variables are binary indicators that capture only whether a worker receives each type of benefit, without information on monetary value. We therefore examine only the extensive margin of benefit provision.

We estimate the following linear probability model for each benefit type:

$$Benefit_{it} = \beta \cdot Outsourced_i + \gamma X_{it} + \delta_{omt} + \varepsilon_{it} \quad (26)$$

where $Outsourced_i$ is an indicator for workers employed by outsourcing firms, X_{it} includes demographic controls (such as education category, gender, race, and age polynomial interactions), and δ_{omt} represents occupation-municipality-time fixed effects.

Table [B.3](#) presents the results. The findings are mixed and do not indicate a consistent negative relationship between outsourcing and the receipt of non-wage benefits. For guards, outsourced workers are 1.7 percentage points less likely to receive housing assistance but 3.0 percentage points more likely to receive transportation assistance. For cleaners, outsourced workers are less likely to receive housing and education benefits, but more likely to receive food and transportation assistance; the effect on health benefits is negative but statistically insignificant. Since data on the intensive margin are unavailable, we cannot determine the relative importance of each type of benefit.

Table B.3: The Effect of Outsourcing on Non-Wage Benefits within the Formal Sector

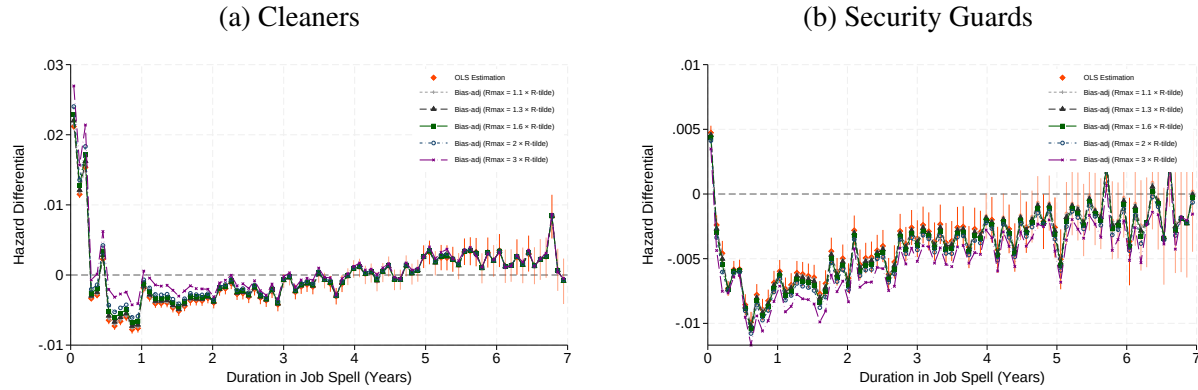
	(1) Housing	(2) Food	(3) Transport	(4) Education	(5) Health
Panel A: Guards					
Outsourced	-0.017*** (0.004)	0.047 (0.028)	0.030* (0.017)	-0.003 (0.002)	0.010 (0.012)
Mean	0.024	0.553	0.683	0.015	0.168
Observations	18,436	18,436	18,436	18,436	18,436
R-squared	0.045	0.137	0.133	0.031	0.109
Panel B: Cleaners					
Outsourced	-0.016*** (0.002)	0.110*** (0.021)	0.142*** (0.021)	-0.007** (0.003)	-0.021 (0.014)
Mean	0.021	0.499	0.604	0.019	0.153
Observations	19,025	19,025	19,025	19,025	19,025
R-squared	0.026	0.102	0.093	0.020	0.052
Demographic Controls	Y	Y	Y	Y	Y
Occ × Year × Microregion FE	Y	Y	Y	Y	Y

Notes: This table presents results from linear probability models estimating the effect of outsourcing on the probability of receiving non-wage benefits. The sample includes formal sector security guards and cleaning workers from the 2003–2010 PNAD surveys. All regressions include demographic controls and occupation-municipality-time fixed effects. *Mean* reports the sample mean of each outcome variable for the full regression sample. Standard errors are clustered at the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C Additional Hazard Results

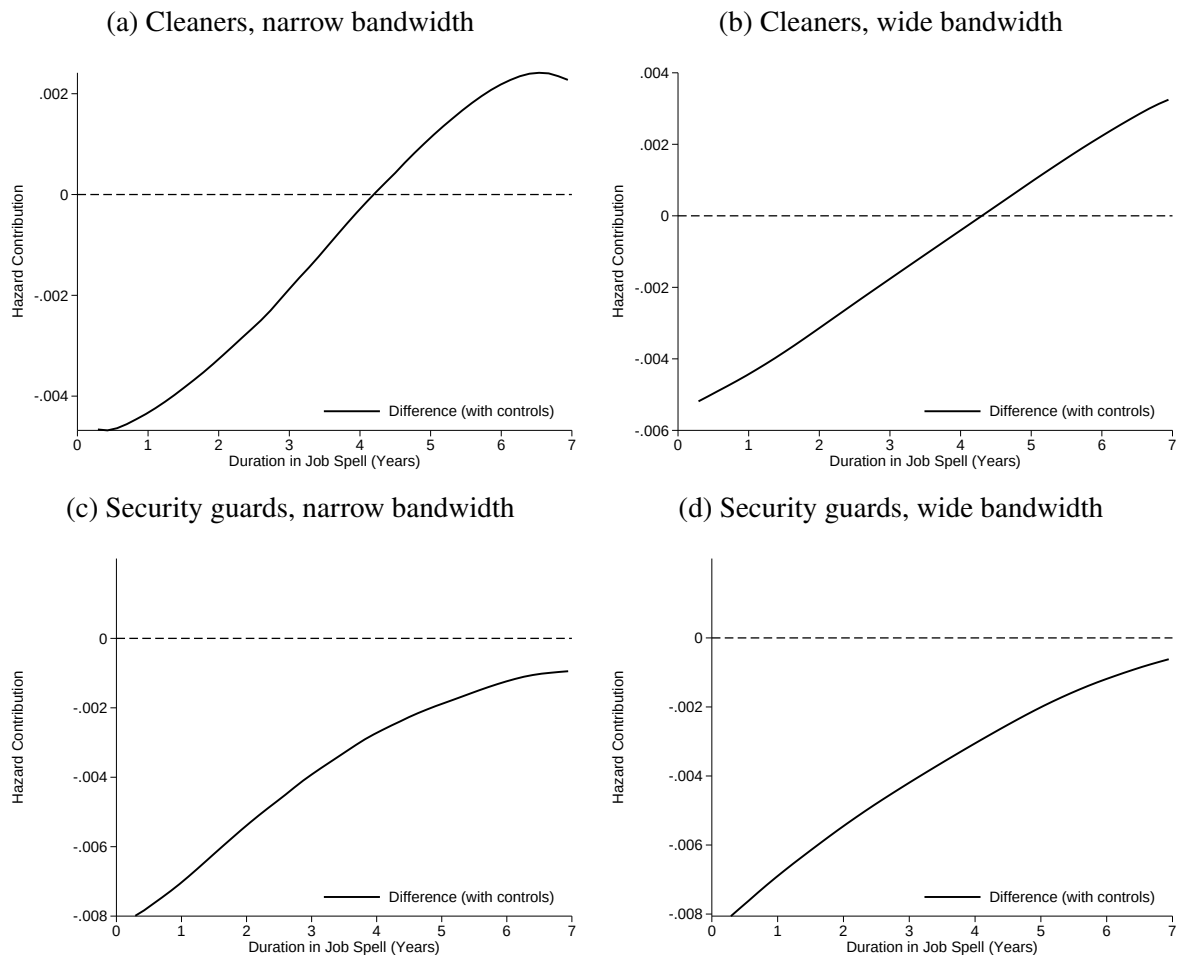
C.1 Robustness of Hazard Estimates

Figure C.1: Oster (2019) Bias-Adjusted EU Hazard Differential under Alternative $R_{\max}$ Values



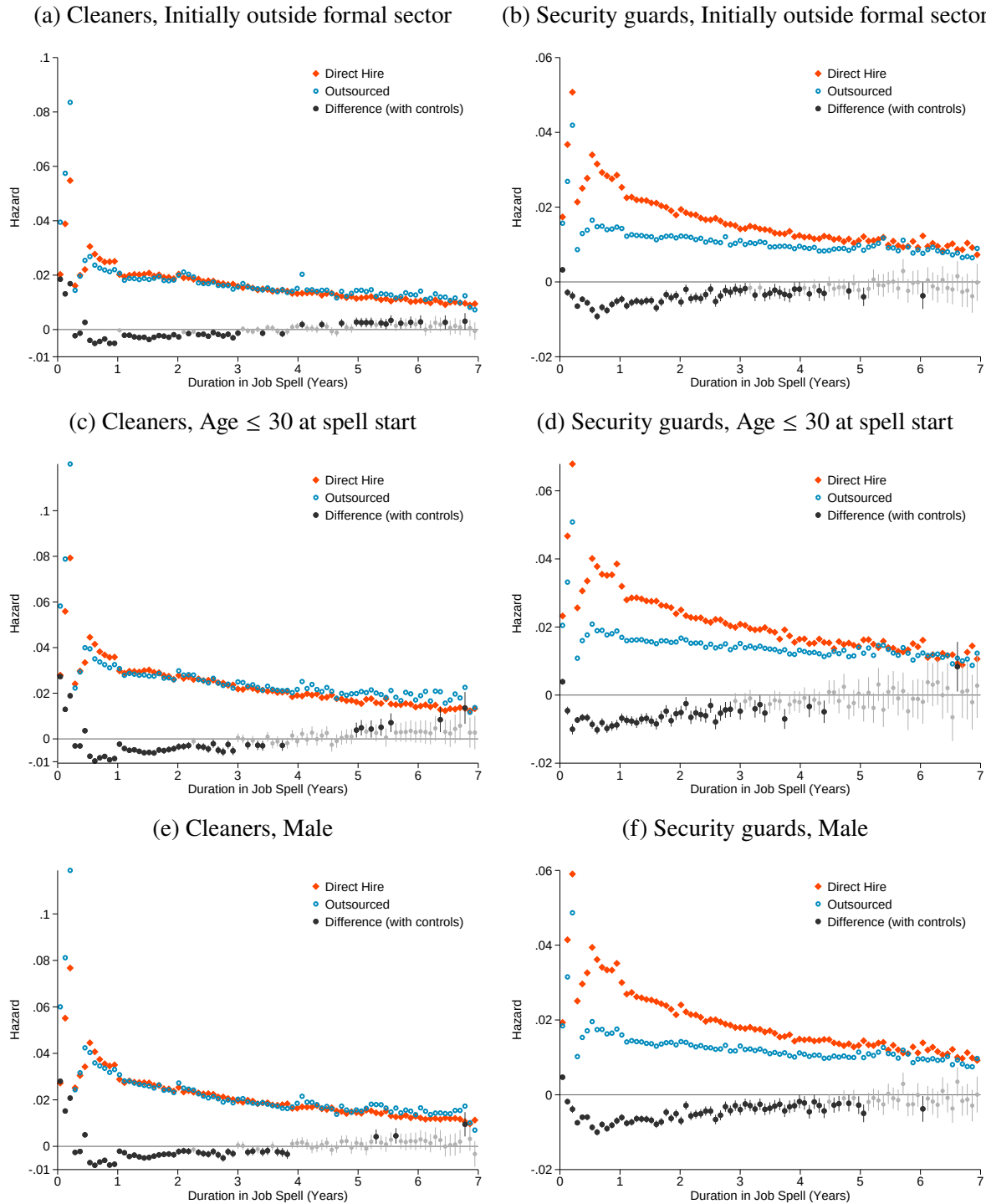
Notes: The figure plots the OLS hazard differential (solid line) alongside the Oster (2019) bias-adjusted coefficients under four values of $R_{\max}$: $1.3\tilde{R}$ (the benchmark), $1.6\tilde{R}$, $2\tilde{R}$, and $3\tilde{R}$. The proportionality parameter is set to $\delta = 1$, meaning unobservable selection is assumed to be as strong as observable selection. We use the full model that controls for sub-occupation by micro-region by starting-year fixed effects, demographic controls, and AKM worker fixed effects.

Figure C.2: Estimates of Hazard from Formal Employment, with Local Linear Smoothing



Notes: This figure plots the estimated hazard differential after linear smoothing using our main specification. The estimated hazard differential from the first three months are dropped, since employment protection legislation applies only after a three-month probationary period. Panels (a) and (c) use local linear smoothing with a bandwidth of 1 year. Panels (b) and (d) use a bandwidth of two years.

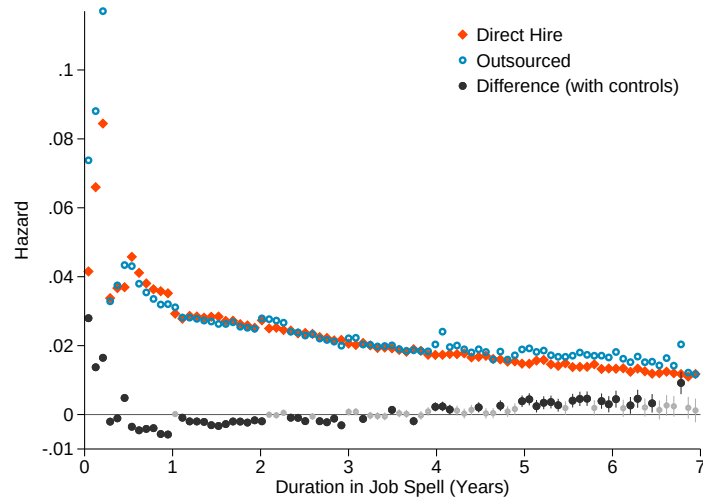
Figure C.3: Effect of Outsourcing on Hazard from Formal Employment, Alternative Samples



Notes: This figure replicates Figure 1 for subsamples. Panels (a) and (b) are restricted to workers who were not employed in the formal sector for at least seven days prior to the beginning of the spell. Panels (c) and (d) are restricted to workers who were age 30 or below. Panels (e) and (f) are restricted to male workers.

Figure C.4: Effect of Outsourcing on Hazard from Formal Employment, Including Quits

(a) Cleaners

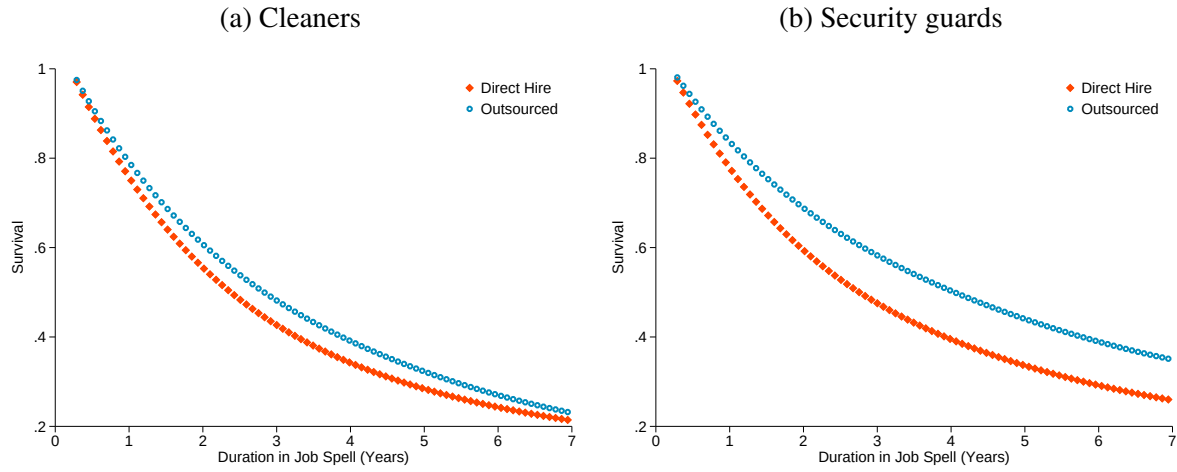


(b) Security guards



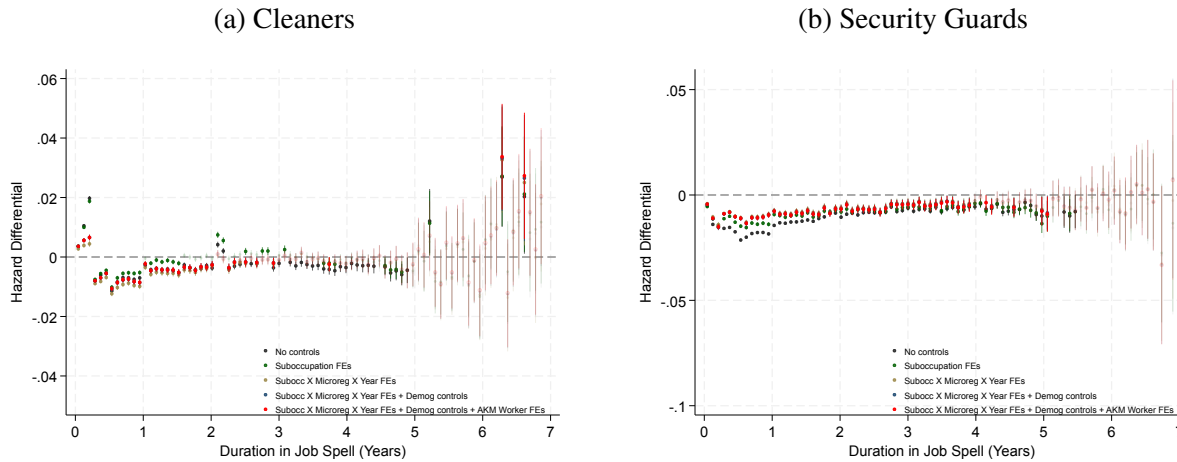
Notes: This figure replicates Figure 1 but does not censor quits.

Figure C.5: Survival Function Based on Estimated Hazard from Formal Employment



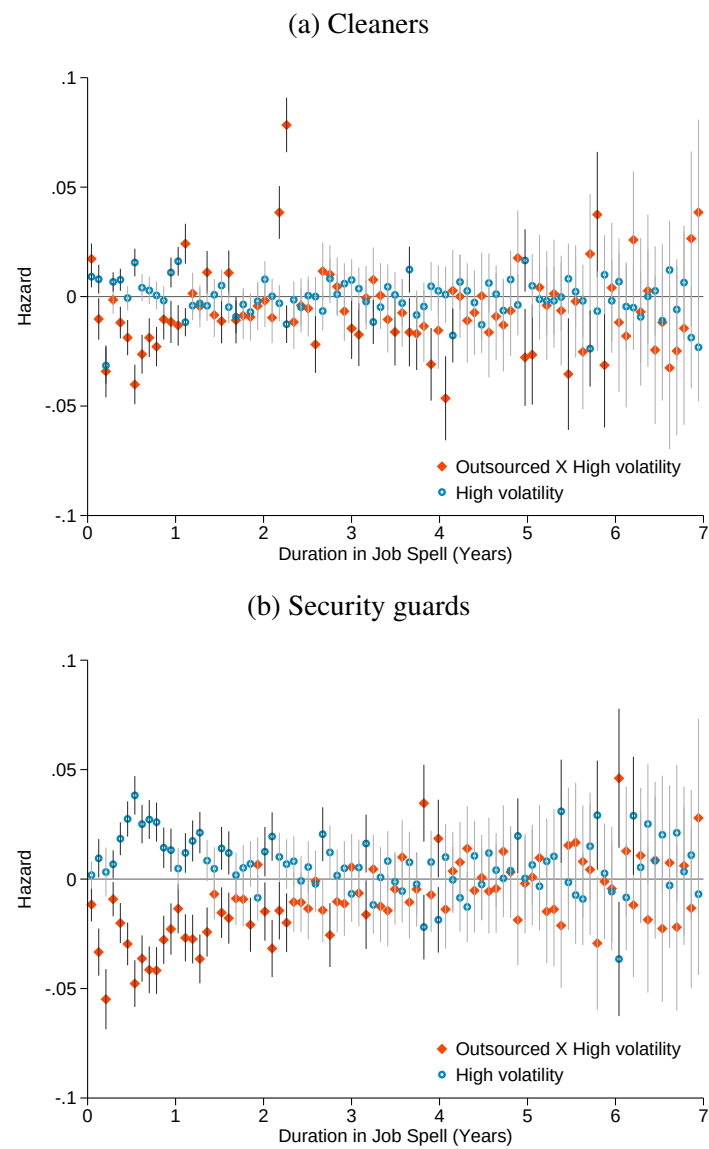
Notes: This figure shows the survival function implied by the hazard estimates from our main specification, after dropping the first three months and smoothing with a bandwidth of one year. The levels are calculated using the predicted mean hazard if observed workers were instead either all outsourced or directly employed.

Figure C.6: EU Hazard Differential under Narrow Outsourcing Definition



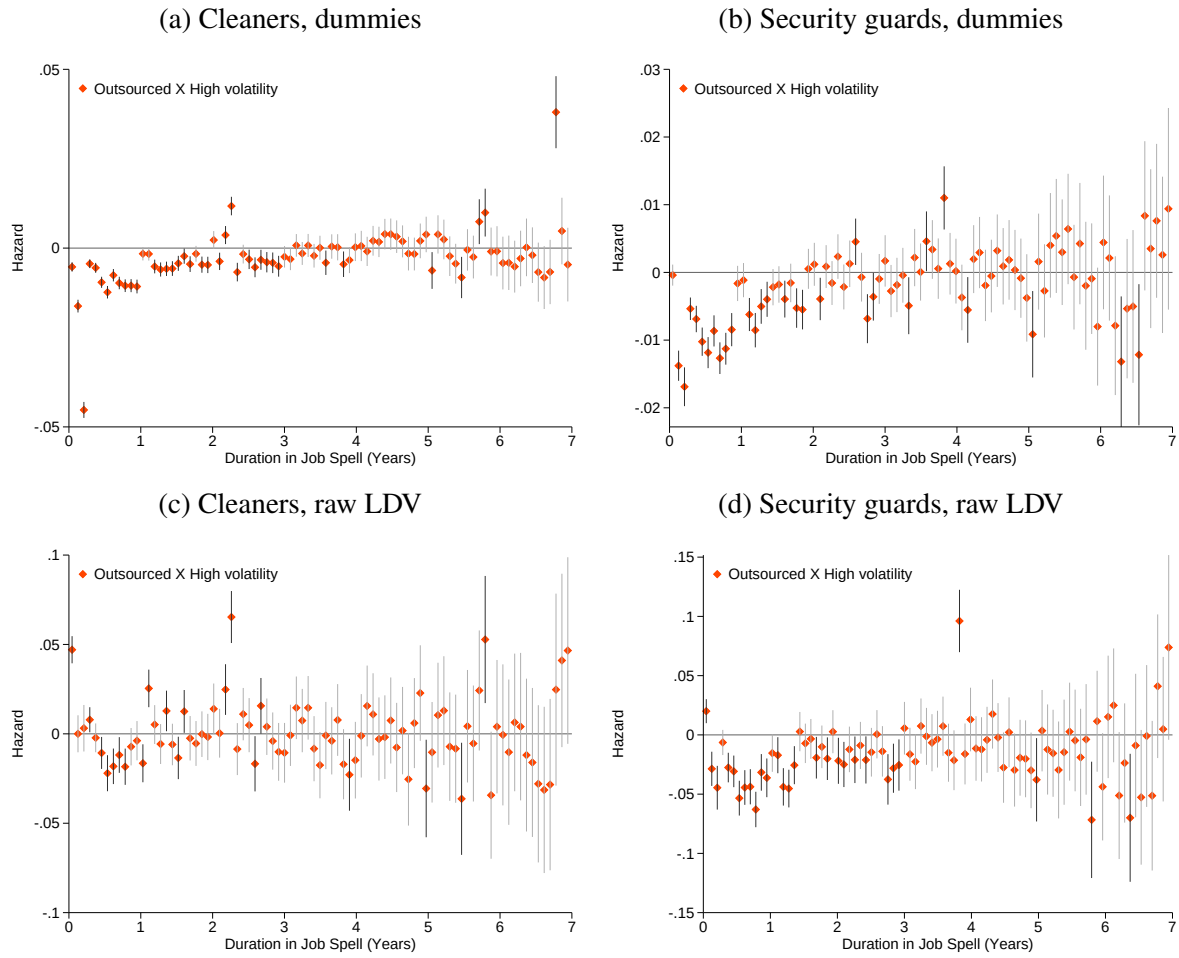
Notes: The figure plots the estimated EU hazard differential from five specifications with increasingly rich sets of controls, using the narrow definition of outsourcing. Specification (1) has no controls; (2) adds sub-occupation FE; (3) adds sub-occupation by micro-region by year FE; (4) adds demographic controls; (5) adds worker fixed effects. The narrow definition restricts outsourced workers to those employed at firms in CNAE 1.0 code 74608 (guards) or 74705 (cleaners).

Figure C.7: Effect of Outsourcing on Hazard from Formal Employment, Interacting with Raw Labor Market Volatility



Notes: This figure replicates Figure 5 but instead uses the raw labor market volatility numbers in each city.

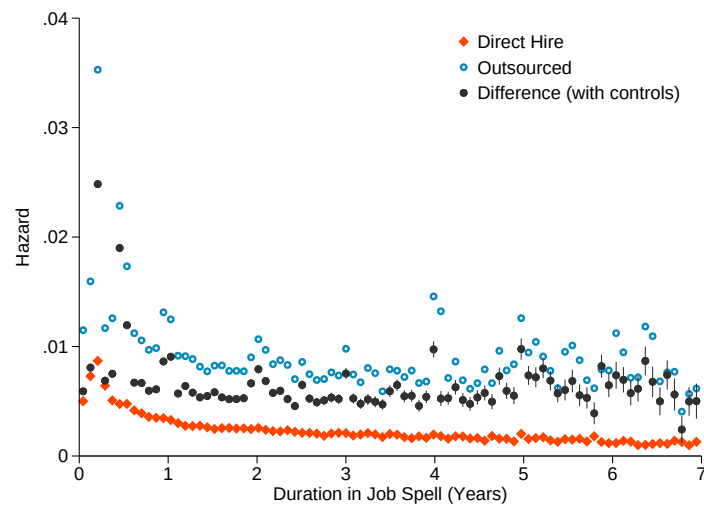
Figure C.8: Effect of Outsourcing on Hazard from Formal Employment, Interacting with Labor Market Volatility, Alternative Controls



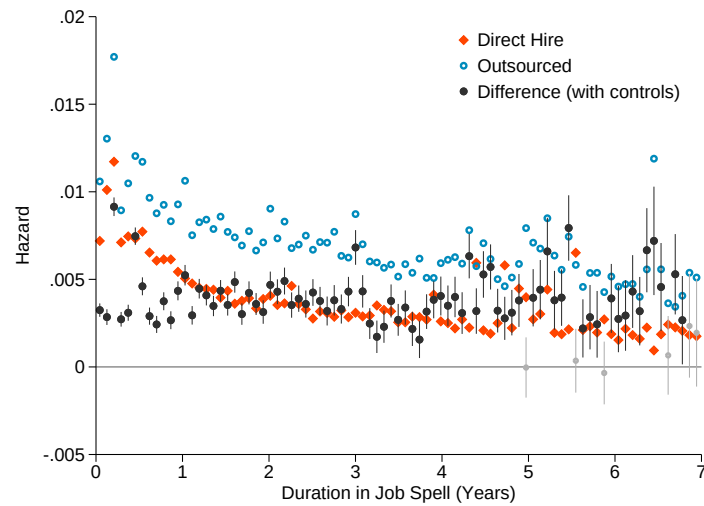
Notes: This figure replicates Figure 5 but controls the suboccupation X microregion X year fixed effects and baseline demographic controls. We only plot the coefficient of the interaction between outsourced and labor market volatility.

Figure C.9: Effect of Outsourcing on Transitions to Other Jobs

(a) Cleaners



(b) Security guards



Notes: This figure replicates Figure 1 but instead shows transitions to other jobs as the hazard outcome.

Table C.1: Employment Status, One Month or Year after Involuntary Exit from Formal Employment

	Cleaners		Security guards	
	Direct-hire	Outsourced	Direct-hire	Outsourced
One month after				
Total formally employed	0.05	0.08	0.05	0.09
Same occupation, direct hire	0.01	0.01	0.01	0.01
Same occupation, outsourced	0.00	0.02	0.01	0.04
Different occupation	0.04	0.05	0.03	0.03
Observations	1341210	845388	386633	400652
One year after				
Total formally employed	0.36	0.38	0.32	0.39
Same occupation, direct hire	0.06	0.04	0.08	0.04
Same occupation, outsourced	0.02	0.08	0.04	0.19
Different occupation	0.29	0.26	0.20	0.16
Observations	1120331	718937	319257	332758

Table C.2: Effects of Outsourcing on Hazard from Formal Employment at year 2, 4 and 6, Alternative Controls

(a) Cleaners					
	(1)	(2)	(3)	(4)	(5)
Effect at 2 years	-0.002 (0.000)	-0.005 (0.000)	-0.004 (0.000)	-0.004 (0.000)	-0.004 (0.000)
R^2	0.000	0.011	0.070	0.071	0.072
Effect at 4 years	0.001 (0.000)	-0.000 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
R^2	0.000	0.013	0.086	0.088	0.088
Effect at 6 years	0.004 (0.001)	0.004 (0.001)	0.004 (0.001)	0.004 (0.001)	0.003 (0.001)
R^2	0.000	0.014	0.088	0.090	0.090
Suboccupation FEs	X				
Subocc X Microreg X Year FEs			X	X	X
Demographic controls				X	X
AKM Worker FEs					X
(b) Security guards					
	(1)	(2)	(3)	(4)	(5)
Effect at 2 years	-0.009 (0.000)	-0.007 (0.000)	-0.006 (0.001)	-0.006 (0.001)	-0.007 (0.001)
R^2	0.001	0.004	0.033	0.034	0.034
Effect at 4 years	-0.004 (0.001)	-0.002 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.002 (0.001)
R^2	0.000	0.003	0.040	0.042	0.042
Effect at 6 years	-0.004 (0.001)	-0.002 (0.001)	-0.003 (0.002)	-0.003 (0.002)	-0.004 (0.002)
R^2	0.000	0.005	0.050	0.053	0.053
Suboccupation FEs	X				
Subocc X Microreg X Year FEs			X	X	X
Demographic controls				X	X
AKM Worker FEs					X

Notes: This table shows the estimation of effects at 2 years, 4 years and 6 years in Figure 3. Different sets of fixed effects and controls are included.

Table C.3: Average Labor Market Volatility in Six Main Cities, 2003-2010

Labor Demand Volatility	
Recife	0.748
Belo Horizonte	0.552
Salvador	0.527
Sao Paulo	0.517
Porto Alegre	0.472
Rio de Janeiro	0.402

Notes: This Table shows the average labor market volatility in Figure 5. Labor market volatility is the average absolute month-to-month change in unemployment rate between 2003 and 2010. The unit of labor market volatility is percentage point.

C.2 Mutual Agreement Separations: Evidence from CAGED

One concern is that workers and firms may have incentives to relabel voluntary quits as involuntary layoffs (e.g., so that workers can receive unemployment benefits). To investigate this, we use CAGED (Cadastro Geral de Empregados e Desempregados) data from 2017 to 2025, which records all hiring and separation events with detailed separation reason codes. Brazil’s 2017 Labor Reform introduced “mutual agreement separations” (Code 90) as a legal alternative to previously informal collusive arrangements. If outsourcing firms were engaging in systematic relabeling, they should be more likely to use this new legal category.

Table C.4 presents descriptive statistics. Mutual agreement separations account for only 2.55% of all separations in our sample. Among worker types, outsourced workers experience mutual agreement separations at a rate of 3.49%, compared to 1.69% for direct-hire workers.

We estimate the following regression model at the separation level:

$$mutual_agreement_{it} = \beta \cdot outsourced_i + \gamma X_{it} + \delta_{omt} + \varepsilon_{it} \quad (27)$$

where $outsourced_i$ is an indicator for workers employed by outsourcing firms, X_{it} includes demographic controls (education category $\times$ gender $\times$ race $\times$ age polynomial interactions), and δ_{omt} represents occupation $\times$ municipality $\times$ time fixed effects.

Table C.5 confirms that workers in outsourcing firms are more likely to experience mutual

Table C.5: Mutual Agreement Separations: Outsourced vs. Direct-hired Workers

	(1) All Workers	(2) Cleaners	(3) Guards
Outsourced	0.003*** (0.001)	0.002** (0.001)	0.010*** (0.002)
Observations	5,509,713	4,142,285	1,367,428
R-squared	0.208	0.158	0.265
Demographic Controls	Y	Y	Y
Occ $\times$ Month $\times$ Microregion FE	Y	Y	Y

Notes: This table reports coefficients from linear probability models where the dependent variable is an indicator for mutual agreement separations. The sample includes security guards and cleaning workers in CAGED separation records (2017–2025), excluding hiring events. Demographic controls include education category $\times$ gender $\times$ race $\times$ age polynomial fixed effects. Standard errors are clustered at the worker level. *** $p < 0.01$, ** $p < 0.05$.

agreement separations, but the magnitude is small. Among all separations, outsourced cleaners (guards) are approximately 0.2 (1.0) percentage points more likely to be involved in mutual agreement than their direct-hired counterparts. These effects are unlikely to substantially bias our main findings.

Table C.4: Descriptive Statistics: Mutual Agreement Separations

Worker Type	Observations	Mean	Std. Dev.
Direct-Hire Workers	2,964,991	0.0169	0.1288
Outsourcing Workers	2,718,371	0.0349	0.1834
Total	5,683,362	0.0255	0.1576

Notes: This table presents descriptive statistics for mutual agreement separations by worker type using CAGED separation records (2017–2025). The sample includes only separation events for security guards and cleaning workers, excluding hiring events. The mutual agreement variable equals 1 if the separation reason is “mutual agreement” (code 90) and 0 for other separation types.

D Proofs of the Model

Proof of Proposition 1

Let $G_a = G(\hat{z}_a)$ and $\eta = 1 - \gamma$. Let $L(t)$ be the ratio between two hazard rates:

$$L(t) \equiv \frac{h_O(t)}{h_E(t)} = \frac{\eta G_O}{G_E} \frac{G_E + (1 - G_E)e^{\lambda t}}{G_O + (1 - G_O)e^{\eta \lambda t}} \quad (28)$$

Let $\xi(t) \equiv \frac{G_E + (1 - G_E)e^{\lambda t}}{G_O + (1 - G_O)e^{\eta \lambda t}}$. We can then write $L(t) = \frac{\eta G_O}{G_E} \xi(t)$.

We first prove two useful lemmas.

Lemma D.1. $L(T) = 1$ if and only if $T = \frac{1}{\lambda} \log \left[\frac{\gamma}{e^{1-\gamma}(1-1/G(\hat{z}_O)) - (1-\gamma)(1-1/G(\hat{z}_E))} \right]$.

Proof. By equation (28), $L(T) = 1$ if and only if

$$e^{\lambda T} = \frac{1 - \eta}{(1 - 1/G_O)e^{\eta} - (1 - 1/G_E)\eta} \quad (29)$$

□

Lemma D.2. $\xi'(t)$ is decreasing in η for all t .

Proof. Note that $\xi(t) > 0$, so $\frac{d \log \xi(t)}{dt} = \frac{\xi'(t)}{\xi(t)}$ has the same sign as $\xi'(t)$. Taking logarithm, we have:

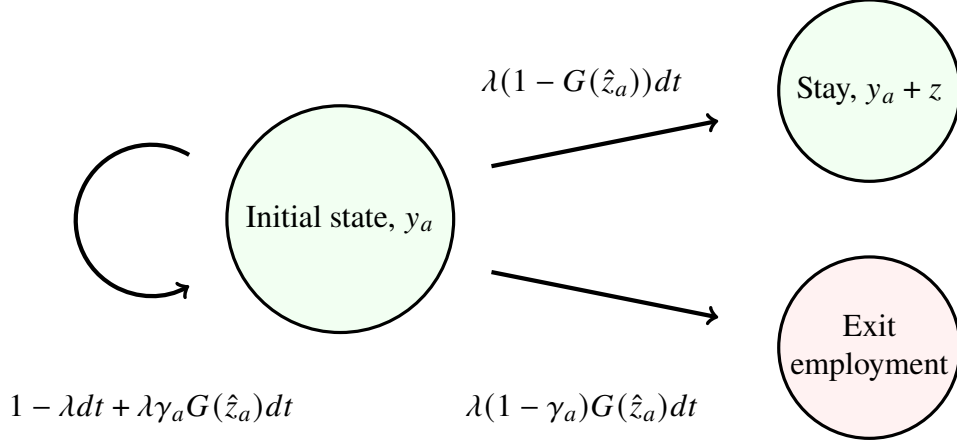
$$\log \xi(t) = \log[G_E + (1 - G_E)e^{\lambda t}] - \log[(G_O + (1 - G_O)e^{\eta \lambda t})] \quad (30)$$

It follows that

$$\frac{d \log \xi(t)}{dt} = \frac{(1 - G_E)\lambda e^{\lambda t}}{G_E + (1 - G_E)e^{\lambda t}} - \frac{(1 - G_O)\eta \lambda e^{\eta \lambda t}}{G_O + (1 - G_O)e^{\eta \lambda t}} \quad (31)$$

Notice that only the second term is related to η . Letting $c = \eta \lambda$, the second term can be rewritten

Figure D.10: Worker State Transitions



as $-\frac{(1-G_O)ce^{ct}}{G_O+(1-G_O)e^{ct}}$. Taking derivative over c and noting that $G_O \in (0, 1)$, we have:

$$-\frac{\partial \frac{(1-G_O)ce^{ct}}{G_O+(1-G_O)e^{ct}}}{\partial c} = -\frac{\overbrace{e^{ct}(G_O-1)[e^{ct}(G_O-1)-G_O(1+ct)]}^{>0}}{\underbrace{[-e^{ct}(G_O-1)+G_O]^2}_{<0}} < 0 \quad (32)$$

If η increases, then $c = \eta\lambda$ increases, and thus $\frac{d\log\xi(t)}{dt}$ decreases. $\square$

The desired proposition follows from combining the following lemmas.

Lemma D.3. *If $\eta = 1$, then $h_O(t) \lesseqgtr h_E(t)$ for all t if and only if $y_O \gtrless y_E$.*

Proof. At $\eta = 1$, we have:

$$\begin{aligned} \frac{d\log\xi(t)}{dt} &= \frac{(1-G_E)\lambda e^{\lambda t}}{G_E + (1-G_E)e^{\lambda t}} - \frac{(1-G_O)\lambda e^{\lambda t}}{G_O + (1-G_O)e^{\lambda t}} \\ &= \frac{(1-G_E)\lambda e^{\lambda t}[G_O + (1-G_O)e^{\lambda t}] - (1-G_O)\lambda e^{\lambda t}[G_E + (1-G_E)e^{\lambda t}]}{[G_E + (1-G_E)e^{\lambda t}][G_O + (1-G_O)e^{\lambda t}]} \\ &= \frac{[G_O(1-G_E) - G_E(1-G_O)]\lambda e^{\lambda t}}{[G_E + (1-G_E)e^{\lambda t}][G_O + (1-G_O)e^{\lambda t}]} \\ &= \frac{(G_O - G_E)\lambda e^{\lambda t}}{[G_E + (1-G_E)e^{\lambda t}][G_O + (1-G_O)e^{\lambda t}]} \end{aligned} \quad (33)$$

The sign of $\frac{d \log \xi(t)}{dt}$ is the same as that of $G_O - G_E$. When $\eta = 1$, $L(0) = \frac{G_O}{G_E}$. If $y_O \geq y_E$, then $G_O \leq G_E$, so $\frac{d \log \xi(t)}{dt} \leq 0$ and thus $\frac{dL(t)}{dt} \leq 0$. Since $L(0) \leq 1$, $h_O(t) \leq h_E(t)$ for all t . $\square$

Lemma D.4. *If $\eta \in \left(\frac{G_E}{G_O}, 1\right)$ and $y_O < y_E$, then $h_O(t) > h_E(t)$ for all t .*

Proof. When $y_O < y_E$, $\hat{z}_O > \hat{z}_E$, $G_O > G_E > 0$, and $\frac{G_E}{G_O} \in (0, 1)$. Notice that $\xi(0) = 1$, so $L(0) = \eta \frac{G_O}{G_E}$. Since $\eta \in \left(\frac{G_E}{G_O}, 1\right)$, $L(0) > 1$. At $\eta = 1$, since $G_O > G_E$, $\frac{d \log \xi(t)}{dt} > 0$ by Equation (33), which implies that $\xi'(t) > 0$. From Lemma D.2, $\xi'(t)$ is decreasing in η , so $\xi'(t) > 0$ for all $\eta \in (0, 1)$. Therefore $L'(t) > 0$. We conclude that $L(t) > 1$, so $h_O(t) > h_E(t)$ for all t . $\square$

Lemma D.5. *If $\eta < \frac{G_E}{G_O}$ and $y_O < y_E$, then there exists T such that $h_O(t) < h_E(t)$ for all $t < T$ and $h_O(t) > h_E(t)$ for all $t > T$.*

Proof. Since $\eta < \frac{G_E}{G_O}$, $L(0) < 1$. Note that $L(t)$ increases in t for all t (by the same logic as the previous proof). By Equation (28), $L(t) \rightarrow \infty$ as $t \rightarrow \infty$. Since $L(t)$ is continuous and monotone, there exists T such that $L(t) < 1$ for all $t < T$ and $L(t) > 1$ for all $t > T$. $\square$

Lemma D.6. *If $\eta < 1$ and $y_O \geq y_E$, then there exists T such that $h_O(t) < h_E(t)$ for all $t < T$ and $h_O(t) > h_E(t)$ for all $t > T$.*

Proof. Since $y_O \geq y_E$, $G_O \leq G_E$. Since $\eta < 1$, $L(0) = \eta \frac{G_O}{G_E} < 1$. By Equation (28), $L(t) \rightarrow \infty$ as $t \rightarrow \infty$. By the continuity of L , there exists some T_1 such that $L(t) < 1$ for all $t < T_1$, and T_2 such that $L(t) > 1$ for all $t > T_2$. By Lemma D.1, there is a unique T such that $L(T) = 1$, so $T_1 = T_2$. $\square$

Proofs for additional results

Lemma D.7. *T increases in y_O , but decreases in λ and y_E .*

Proof. This follows from Equation (15) and noting that $G_a = G(-y_a + r\bar{W})$. $\square$

Lemma D.8. *T increases in γ if and only if $e^{1-\gamma}(1+\gamma) > \frac{G(\hat{z}_O)}{1-G(\hat{z}_O)} \cdot \frac{1-G(\hat{z}_E)}{G(\hat{z}_E)}$.*

Proof. Equation (29) implies that

$$\frac{de^{\lambda T}}{d\eta} = \frac{(1 - 1/G_O)e^\eta(\eta - 2) + (1 - 1/G_E)}{((1 - 1/G_O)e^\eta - (1 - 1/G_E)\eta)^2}. \quad (34)$$

Therefore, $\frac{dT}{d\gamma} \geq 0$ if and only if

$$e^{1-\gamma}(1 + \gamma) \leq \frac{(1 - 1/G_E)}{(1 - 1/G_O)}. \quad (35)$$

□

E Additional Details for Calculating the NPV

This section provides additional details for calculating the net present value (NPV). Our approach integrates insights from multiple datasets to construct a comprehensive view of career trajectories following separations, including both E-U and E-E transitions. We begin by using PNADc data to estimate transition probabilities, illustrating where workers move after layoffs. Next, we quantify wage changes across different employment states utilizing the panel structure of the same dataset. Finally, we incorporate these findings into our main RAIS event study through an improved wage imputation methodology that accounts for both transition patterns and wage differentials.

E.1 Employment State Transition Analysis

Data: PNADc We utilize PNADc (Pesquisa Nacional por Amostra de Domicílios Contínua) panel data from 2012 to 2019 to analyze transitions from formal to informal employment. While RAIS provides comprehensive administrative records for formal sector workers, it does not capture workers who transition to informal employment or unemployment following layoffs. In contrast, PNADc, a household survey conducted quarterly since 2012, encompasses all employment states—including informal work—and is representative of the Brazilian labor market (Ulyssea, 2010).

The panel structure of PNADc is particularly valuable for our analysis. It tracks the same individuals across quarters, enabling us to observe actual transitions between employment states. Additionally, PNADc’s detailed occupation coding allows us to identify the same worker types—such as outsourced cleaners, direct-hire cleaners, outsourced guards, and direct-hire guards—that we study in RAIS, ensuring consistency in our analysis.

Transition Matrix Results Table E.6 displays the conditional transition probabilities from each of the four worker types to either unemployment or informal employment, based on quarterly panel observations from PNADc (2012–2019).

Table E.6: Transition Probabilities to Unemployment and Informal Employment

Previous State	Informal Sector (%)	Unemployed (%)
Outsourced Guard	67.2	32.8
Direct-Hire Guard	71.5	28.5
Outsourced Cleaner	62.1	37.9
Direct-Hire Cleaner	65.6	34.4
Other Formal Sectors	76.8	23.2

Notes: This table shows the conditional probability of transitioning from each worker type to unemployment or informal employment, based on PNADc panel data (2012–2019). Probabilities are conditional on leaving formal employment and sum to 100% across the two destination states for each worker type. The transition probabilities are calculated using quarterly observations where workers are observed transitioning out of formal sector employment.

The table above highlights significant differences in the likelihood of transitioning to informal employment following layoffs from the formal sector across worker types. Notably, among the four worker types of interest, direct-hire guards have the highest probability of moving into informal work at 71.5%, whereas outsourced cleaners have the lowest at 62.1%.

E.2 Wage Differential Analysis

Data Sample To quantify the wage changes associated with different employment transitions, we utilize PNADc panel data (2012–2019) to estimate wage differentials between formal and informal sector employment. For each formal employment category, we construct samples comprising workers who have been observed in that specific formal sector job at least once during the panel period and who are also observed in informal employment at some point. This restriction ensures that our wage differential estimates are based on workers with relevant occupational experience and allows us to employ individual fixed effects to control for unobserved, time-invariant characteristics.

Regression Specification For each formal employment category a , we estimate the following equation with individual fixed effects:

$$\ln(wage_{it}) = \gamma_a \cdot informal_{it} + \alpha_i + \lambda_t + X_{it}\beta + \varepsilon_{it} \quad (36)$$

where $informal_{it}$ is an indicator for informal employment, α_i are individual fixed effects,

and λ_t are time fixed effects. Following standard Mincer equation specification, we also control for age and age squared, and years of education. Standard errors are clustered at the individual level.

Wage Differential Results Table E.7 displays the estimated wage gap between informal and formal employment, separately for each job type.

Table E.7: Wage Differentials between Informal and Formal Employment

	(1) Outsourced Guard	(2) Direct-Hire Guard	(3) Outsourced Cleaner	(4) Direct-Hire Cleaner	(5) Other Formal Sectors
Informal Penalty	-0.125*** (0.008)	-0.121*** (0.007)	-0.114*** (0.008)	-0.124*** (0.004)	-0.151*** (0.003)
Observations	240,705	232,016	228,588	288,032	315,125
R-squared	0.814	0.811	0.805	0.791	0.8052
Worker FE	Y	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y	Y
Demographic Controls	Y	Y	Y	Y	Y

Notes: This table reports coefficients from fixed-effects regressions of log wages on employment state indicators, following standard Mincer equation specification. Each column represents a separate regression comparing informal sector employment to the specified formal sector job type (the omitted base category). The reported coefficients thus represent the wage penalty associated with informal employment relative to each formal occupation. The sample in each column is restricted to worker-quarter observations where the individual is employed in either the informal sector or the specified formal occupation. All regressions include worker fixed effects and quarter fixed effects. Controls include age, age squared, years of education, gender, and race. Standard errors are clustered at the individual level. *** p<0.01.

The results show a consistent wage penalty for informal employment across job types. Workers in the informal sector earn 11 to 12 percent less compared to their counterparts in the formal sector. This finding is consistent with [Meghir, Narita and Robin \(2015\)](#), who document substantial wage differentials between formal and informal employment in Brazil.

E.3 Wage Effect of E-U and E-E Transitions

We now integrate the transition probabilities and wage differentials estimated from PNADc to impute wages for workers transitioning out of formal employment. Subsequently, we conduct

event studies to analyze the wage effects of E-U and E-E transitions.

Sample Construction Our sample is derived from RAIS administrative records, covering security guards and cleaning workers from 2003 to 2010. For each worker type – outsourced guards, direct-hire guards, outsourced cleaners, and direct-hire cleaners – we construct separate analytical samples following a consistent methodology.

First, we identify all workers who have ever been employed in the target occupation during the sample period. Second, we create a balanced worker-year panel that includes all years from each worker’s first to last observation, filling in years when workers are not observed in formal employment.

Next, we employ matching techniques to construct comparable treatment and control groups. The matching is performed separately for each event cohort (defined by the year of the employment-to-unemployment or employment-to-employment event) and for each worker type.

For a given event year y , the treatment group includes workers who experience their first event in that year. The control pool comprises workers who either experience their first event in a later year (not-yet-treated) or never experience the event during the sample period (never-treated). Importantly, we ensure that control workers are employed in the same occupation as treated workers at $t = -1$, which allows for meaningful comparisons between similar worker types.

We begin by requiring exact matching on the worker’s geographic region at $t = -1$ and their employment history during the two years prior to the event (at $t = -1$ and -2).¹⁸ Next, we select a control worker with the closest match to each treated worker based on the following dimensions measured at $t = -1$: age, years of schooling, and wages at $t = -1$ and -2 . Using a nearest neighbor approach, we assign one control worker to each treated worker. This combination of distance-based and exact matching ensures that treated and control workers are highly comparable in terms of observable characteristics, local labor market conditions, and recent labor market trajectories.

¹⁸It is important to note that we can only identify workers employed in the formal sector and cannot distinguish between unemployed individuals and those in the informal sector.

Outcome Variables We examine two outcome variables that together offer a comprehensive view of post-separation labor market outcomes. First, we analyze the impact on **imputed log wage**, which estimates wages for periods when workers are not observed in formal employment within RAIS. The imputation process involves two steps.

First, for any year a worker is missing from RAIS, we carry forward their last observed formal employment status (e.g., Outsourced Guards) and their last observed wage. If the worker’s last formal employment does not fall into one of the four main categories – outsourced guard, direct-hire guard, outsourced cleaner, or direct-hire cleaner – they are classified as belonging to “Other Formal Sectors.”

We then calculate the expected wage when outside the formal sector. This is a weighted average of the informal wage and zero (representing unemployment). The expected informal wage is estimated by adjusting the last observed formal wage $w_{i,\tau}$ by the formal-informal wage differential γ_a specific to worker type a . The final imputed wage is given by:

$$\hat{w}_{it} = P_{a \rightarrow informal} \times (w_{i,\tau} \times e^{\gamma_a}) + (1 - P_{a \rightarrow informal}) \times 0 \quad (37)$$

where $w_{i,\tau}$ is the last observed wage at time τ , $P_{a \rightarrow informal}$ is the probability of transitioning to informal employment for worker type a , and γ_a is the corresponding informal-formal log wage differential. If a worker has no prior formal employment history in the panel, their missing wages are imputed as zero.

Additionally, we use **original wage** as an outcome variable as robustness checks. To address potential issues with zero wages for workers not formally employed, we use the wage ratio, which compares a worker’s current wage to their wage at $t = -1$.

Regression Specification Our event study specification employs a staggered difference-in-differences design with the following structure:

$$Y_{it} = \sum_{\tau \neq -1} \delta_{EU}^a(\tau) \cdot D_{it}^\tau \cdot T_i + \sum_{\tau \neq -1} \theta_\tau \cdot D_{it}^\tau + \xi \cdot T_i + \alpha_c + \lambda_t + \varepsilon_{it} \quad (38)$$

where Y_{it} is the wage outcome for worker i in year t ; T_i is an indicator for belonging to the

treatment group; D_{it}^τ is an indicator for being τ years from the event (with $\tau = -1$ as the omitted reference period); α_i are worker fixed effects; and λ_t are calendar year fixed effects.

The coefficients of interest are $\delta_{EU}^a(\tau)$, which capture the dynamic treatment effects – the difference in outcomes between treated and control workers at each time period relative to the event, compared to the difference at $t = -1$. The pre-event coefficients ($\delta_{EU}^a(\tau)$ for $\tau < -1$) provide a test of the parallel trends assumption: if treated and control workers were on similar trajectories prior to the event, these coefficients should be close to zero. Standard errors are clustered at the worker level to account for serial correlation within individual workers’ outcomes over time.

Event study results for E-U transitions The event study for the E-U transition reveals a significant scarring effect of layoffs, with notable differences across job types. As shown in Figure 6, following an involuntary separation at $t = 0$, all groups experience sharp, immediate declines in both log imputed wages and the wage ratio. Specifically, there is an immediate drop in imputed wages of approximately 50% for outsourced and direct-hire guards, and about 55% for outsourced and direct-hire cleaners.

However, outsourced workers recover more rapidly and maintain this advantage in the long term. By $t = +5$, five years after the layoff, the wage loss (measured via imputed wages) for outsourced guards is 18%, compared to a much larger 35% for direct-hire guards. For cleaners, outsourced workers experience a 16% wage loss after five years, whereas their direct-hire counterparts face a more substantial loss of 33%. This suggests that the flexible nature of the outsourcing market enables displaced workers to re-enter formal employment more quickly, thereby reducing long-term unemployment spells and persistent wage scars.

Figure E.11 presents results using the original wages, revealing a similar pattern but with substantially larger wage losses. This is likely because, without accounting for the fact that some of the “unemployed” workers are still employed in the formal sector, the measured wage losses appear more severe.

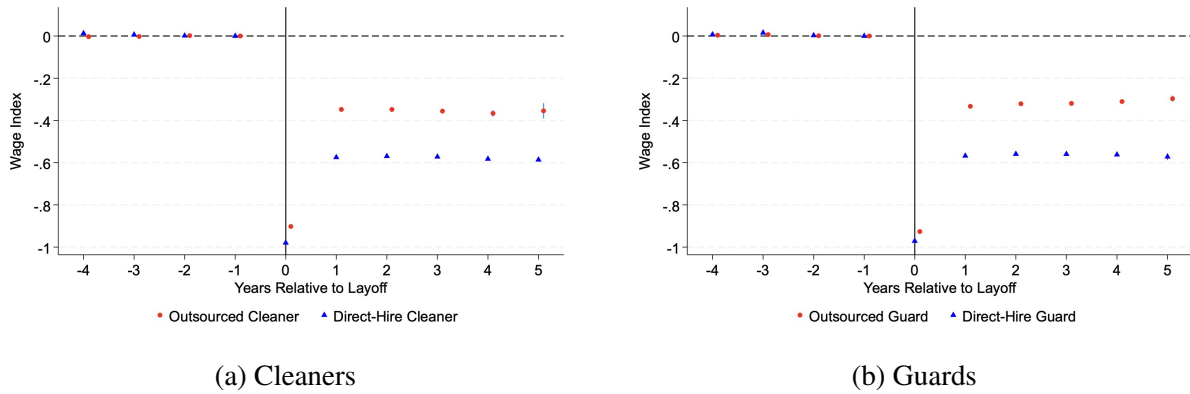
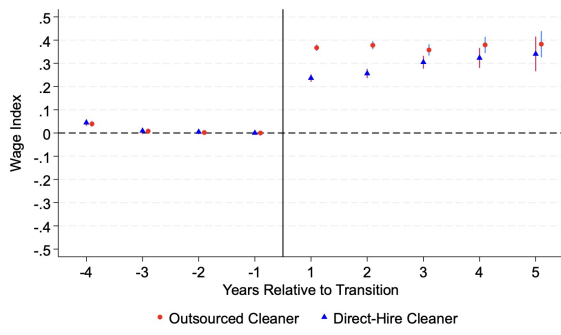


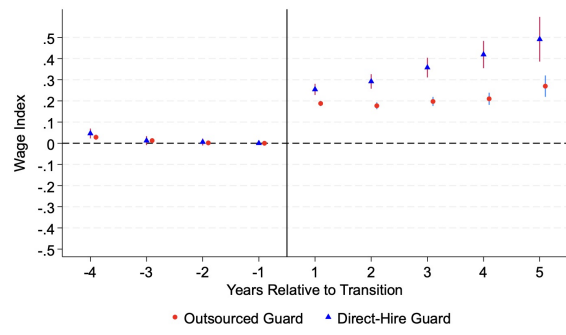
Figure E.11: Wage Effects of E-U Transition (Original Wages)

Event study results for E-E transitions Figure 7 depicts the impact of E-E transitions on imputed wages, following the same specification as in Equation 38. For cleaners, the transition leads to an average wage increase of 27% within the first five years for outsourced workers, which exceeds the 19% increase observed for direct-hire workers. Conversely, among guards, the pattern is reversed: direct-hire guards experience a higher wage increase of 21%, compared to 15% for outsourced guards.

Figure E.12 shows the event study results using the original wages. The observed wage increases following E-E transitions are larger when using original wages than with imputed wages. This likely occurs because E-E workers are more likely to remain employed in the formal sector compared to their control counterparts. Nevertheless, the findings still indicate that outsourced cleaners experience greater wage growth than their direct-hire counterparts, while for guards, the wage growth is higher for direct-hire workers than for outsourced workers.



(a) Cleaners



(b) Guards

Figure E.12: Wage Effects of E-E Transition (Original Wages)